# Financial Conditions, $r^*$, and the Transmission of Monetary Policy Surprises

Draft for discussion

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**Abstract**

Monetary policy is the main tool for stabilizing the economy, but estimates of its effects have been unreliable. In samples that start in the mid-1990s, standard methods applied to surprises around Federal Reserve announcements give unemployment that falls after a tightening and prices that do not respond. I keep the surprises and change how their effects are carried through the economy. I measure how the two-year yield and a broad financial conditions index move in the half hour around each announcement, and carry those moves forward in a daily and a monthly model. Both are measured relative to the neutral rate. Carried through the two-year yield alone, the surprises give the familiar wrong signs. Carried through the index relative to the neutral rate, a tightening raises unemployment and lowers the price level, and the ratio of the price response to the unemployment response is of conventional size. The main results hold across alternative samples, measures of the neutral rate, trends, credit conditions and surprises.

## 1 Introduction

Monetary policy is the main tool for stabilizing the economy. Its use rests on estimates of how much a change in policy moves unemployment and inflation. In samples that start after the early 1980s, standard methods often give a surprise tightening small effects on unemployment, or effects of the wrong sign (Ramey, 2016). Two concerns follow. One is that policy has become less powerful (Boivin and Giannoni, 2006; Boivin et al., 2010). The other is that inflation responds little to slack, so that lowering inflation requires a large and lasting rise in unemployment (Hazell et al., 2022; Del Negro et al., 2020). I ask whether these findings reflect how policy works or how its effects are measured.

The question matters in practice. From 2022 to 2023 the Federal Reserve raised its policy rate by more than 5 percentage points. Policymakers expected costs: in August 2022 the Chair warned of "some softening of labor market conditions" and called these "the unfortunate costs of reducing inflation" (Powell, 2022). They also described policy as working through broad financial conditions: in November 2023 the Chair noted that "Financial conditions have tightened significantly in recent months, driven by higher longer-term bond yields, among other factors" (Powell, 2023). The standard tools for measuring the effects of policy leave those conditions out.

The answer is that measurement matters. I use the policy-news surprise of Nakamura and Steinsson (2018) around 252 FOMC announcements from 1996 to 2026, and I change only how its effects are carried through the economy. Carried through the two-year yield alone, the surprises give the result of the standard methods: unemployment falls after a tightening, by 0.95 percentage point after a year in a recursive VAR and 0.91 in a proxy VAR, and prices do not respond. Carried through a broad financial conditions index, with the index and the yield both measured relative to the neutral rate $r^*$, a policy move that raises the two-year yield 1 percentage point raises

unemployment by 0.50 percentage point after two years and lowers the core price level by 0.81 percent. Lowering the price level 1 percent costs about 0.9 percentage-point years of unemployment (one point of unemployment for one year), a conventional sacrifice ratio. These responses have the signs that theory predicts, where the standard methods give the wrong ones (Figure 1). In the terms of Barnichon and Mesters (2021), the ratio of the price response to the unemployment response is −0.90 after three years, against about −0.15 for 1990–2008 in their estimates.

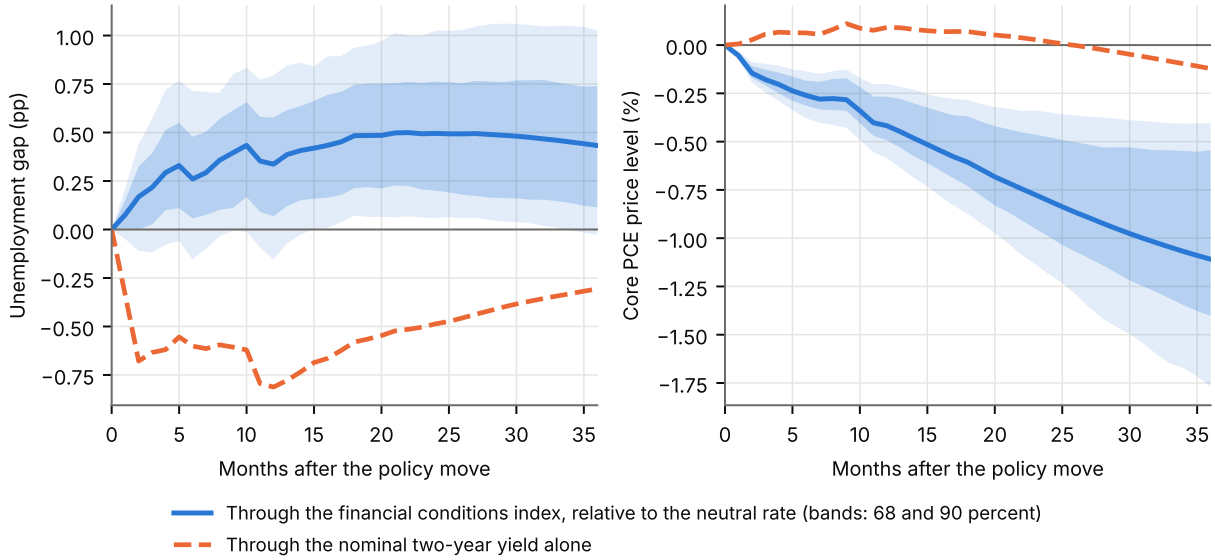



**Figure 1:** Responses to a policy move that raises the two-year yield 1 percentage point in the 30-minute window around the announcement. Solid line: the surprise carried through the financial conditions index and the two-year yield, both measured relative to the neutral rate (shaded bands: 68 and 90 percent intervals). Dashed line: carried through the nominal two-year yield alone. 1996–2026, 12 monthly lags.

The approach has three features.

First, I measure the effect of each surprise in a 30-minute window around the announcement, in which unemployment and inflation cannot move. A daily model then traces how the two-year yield and financial conditions evolve after each announcement, and monthly unemployment and core inflation respond to their monthly averages, as in a standard vector autoregression (VAR). As in Jacobson et al. (2026), the model takes into account when each announcement falls within its month.

Second, financial conditions enter as a whole (Boivin et al., 2010). The financial conditions index (FCI) combines the 10-year yield, equity valuation and the real dollar, weighted by their responses to policy news. A policy surprise raises the FCI about 1 unit per percentage point on the two-year yield. An ordinary monthly rise of 1 percentage point in the two-year yield comes with a rise of only 0.36 to 0.51 unit. A model with the yield alone therefore traces the ordinary move in the yield, which carries much less of the change in financial conditions than a policy surprise does.

Finally, I measure the yield and the index relative to $r^*$, the rate that neither speeds up nor slows down the economy. The neutral rate fell from the 1990s to the 2010s (Laubach and Williams, 2003; Holston et al., 2017, 2023; Hamilton et al., 2016; Bauer and Rudebusch, 2020; Kiley, 2020b), and interest rates trended down with it while activity trended up, so a model that links the level of rates to activity partly picks up these opposite trends. Inflation depends on how long real rates stay above or below $r^*$, so the adjustment should matter most for prices. The FCI and the three

versions (nominal, real, then relative to $r^*$) come from a companion note on financial conditions and were fixed before this analysis (FCI note, 2026).

Earlier fixes improve the surprise or the timing of the outcome. I leave both alone and change the channel: policy reaches the economy through long rates, stock prices and the dollar, and these have to be measured relative to $r^*$. The main results hold across alternative samples, measures of $r^*$, trends, credit conditions and surprises (Section 4).

### 1.1 Connection to the literature

**Surprises and their effects.** Interest-rate surprises around FOMC announcements were meant to settle the measurement problem: the decision precedes the announcement, so market moves in a narrow window around it cannot have caused it (Kuttner, 2001; Gürkaynak et al., 2005). They measure the effects of policy on asset prices precisely (Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018; Swanson, 2021), but for unemployment and inflation the old puzzles often return. Four explanations come with fixes. Announcements reveal the Fed's view of the economy, and separating it gives larger price effects (Romer and Romer, 2000; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021). Part of each surprise is predictable, and removing it sharpens the estimates (Bauer and Swanson, 2023a,b). Markets react in minutes while unemployment and inflation are monthly averages, and measuring inflation daily removes most of the wrong sign (Jacobson et al., 2026). And credit costs, stock prices and the dollar carry part of the effect (Gertler and Karadi, 2015; Caldara and Herbst, 2019; Kroencke et al., 2021; Bauer et al., 2023; Boehm and Kroner, 2024; Bauer et al., 2026; Boehm et al., 2026). Most of this work traces the surprise through a single nominal interest rate. The choice of surprise series (Bu et al., 2021; Brennan et al., 2025) and of lags (Coibion, 2012) can matter. Futures-based surprises in monthly models go back to Faust et al. (2004), and related work on timing and mixed frequencies includes Kilian (2024); Chudik and Georgiadis (2022); Francis et al. (2011); Ghysels (2016); Gazzani and Vicondoa (2020); Chudik and Kilian (2026).

**What is new.** My paper is closest in spirit to Jacobson et al. (2026). Like them, I hold the surprise and the method fixed and change one measurement. They change how often the outcome is measured; I change how the channel from policy to the economy is measured, and my main result is for activity. Bauer and Swanson (2023a) and Jarociński and Karadi (2020) improve the surprise itself. In my design a surprise enters only through its immediate effect on the two-year yield and the FCI and through its timing, so their fixes and mine can in principle be combined.

**Financial channels and risk appetite.** Gertler and Karadi (2015) and Caldara and Herbst (2019) are closest in emphasizing financial channels. I use a broad index measured relative to $r^*$, and the baseline leaves credit spreads out. Studies of the "non-yield" shock treat policy-driven moves in stock prices and the dollar as a second shock (Bauer et al., 2026; Boehm and Kroner, 2024; Boehm et al., 2026); I treat them as part of financial conditions, and keep only the part of those moves that comes with interest-rate news. The response of stock prices to long rates has also changed over time (Kiley, 2014). Caballero et al. (2026) propose a neutral level of the FCI itself, FCI$^*$. I keep $r^*$ for the rates inside the two-year yield and the index; gaps to FCI$^*$ are a natural extension.

**The power of policy and the Phillips curve.** The effects of policy shocks became smaller after the early 1980s, which the literature attributes to a more systematic policy response and better-anchored expectations (Boivin and Giannoni, 2006; Boivin et al., 2010; Primiceri, 2005), or to a failure of standard identification once policy became more forward-looking (Barakchian and Crowe, 2013). Recent work finds a small Phillips-curve slope (Hazell et al., 2022; Del Negro et al., 2020; Ball and Mazumder, 2019; Fitzgerald et al., 2024), hard to identify when policy offsets

demand shocks (McLeay and Tenreyro, 2020) and understated by conventional methods (Barnichon and Mesters, 2020). Policy shocks offer one way to identify it. Barnichon and Mesters (2021) measure the tradeoff by the ratio of cumulative inflation to cumulative unemployment after such shocks and find it much flatter after 1990 than before. I add evidence of this kind and show that the measured tradeoff depends on how the stance of policy is measured.

## 2 Empirical approach

The model has three pieces. A surprise moves the two-year yield and the FCI by amounts measured in the window: the vector $b$ in equation (10). A daily VAR, equation (11), carries those two series forward. Their monthly averages then enter a monthly VAR for unemployment and inflation, equation (12). Appendix F gives the construction of the series, the algorithm for tracing responses and the bootstrap, and Appendix G lists the data.

### 2.1 Measuring financial conditions

The FCI is the market index of FCI note (2026): a weighted sum of the 10-year Treasury yield, stock valuations and the real broad dollar. Indexes of this kind are weighted sums of financial variables (Hatzius et al., 2010). They differ mainly in where the weights come from. FCI-G takes them from large macroeconomic models (Ajello et al., 2023). Mine come from the event-study regressions long used to measure how asset prices respond to policy (Kuttner, 2001; Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018).

**Policy news.** Let $\Delta X_a$ be the $5 \times 1$ vector of changes in the current- and next-meeting federal funds futures and the second to fourth eurodollar futures in the window around announcement $a$. Following Nakamura and Steinsson (2018), let $p_a$ be the first principal component of these changes, signed so that it moves with the fourth eurodollar future. Policy news is $p_a$ rescaled so that one unit moves the two-year yield $y^2$ by 1 percentage point:

$$s_a = \gamma\, p_a, \qquad \gamma = \frac{\sum_a p_a\, \Delta y_a^2}{\sum_a p_a^2}. \tag{1}$$

**Weights.** Let $\Delta A_a$ be the vector of changes in the 10-year yield, the S&P 500 and the dollar index (DXY) in the window. The companion note treats these changes as

$$\Delta A_a = \psi\, s_a + \eta_a, \tag{2}$$

where $\psi$ holds the policy responses and $\eta_a$ is other news, with covariance matrix $\Sigma$. It estimates $\psi$ by regressing each element of $\Delta A_a$ on $s_a$ and a constant, and $\Sigma$ from the residuals, over 253 announcements from January 1996 to September 2026. The weights are those that recover the policy component most precisely (one-factor generalized least squares):

$$w = \frac{\Sigma^{-1}\psi}{\psi' \Sigma^{-1} \psi}. \tag{3}$$

Because $w'\psi = 1$ and $s_a$ is scaled to the two-year yield, a policy move that raises the two-year yield 1 percentage point raises the index by 1 unit. The weights are 0.61 on the 10-year yield, $-0.051$ per percent on stock valuations and 0.065 per percent on the dollar. Each of the three accounts for about a third of the index's response to policy.

**Three versions.** In month $t$, the index in version $v$ is

$$F_t^v = w_B\, B_t^v + w_Q\, Q_t^v + w_D\, D_t, \tag{4}$$

measured as a deviation from its sample mean. Here $D_t$ is 100 times the log of the real broad dollar in every version. With $y_t^{10}$ the 10-year yield, $\pi_t^{10}$ 10-year expected inflation, and $(E/P)_t$ the inverse of the cyclically adjusted price-earnings ratio (CAPE), the other two terms are:

$$\text{Nominal:} \quad B_t = y_t^{10}, \qquad Q_t = 100 \ln \text{CAPE}_t, \tag{5}$$

$$\text{Real:} \quad B_t = y_t^{10} - \pi_t^{10}, \qquad Q_t = 100 \ln \text{CAPE}_t, \tag{6}$$

$$r^*\text{-gap (baseline):} \quad B_t = y_t^{10} - \pi_t^{10} - r_t^*, \qquad Q_t = -\frac{100}{\overline{E/P}} \left[ (E/P)_t - r_t^* \right]. \tag{7}$$

In the $r^*$-gap version, stocks are measured by the gap between their earnings yield and $r^*$. The term $100/\overline{E/P}$, with $\overline{E/P}$ the sample average, converts this gap to the units of the log CAPE (the cyclically adjusted price-earnings ratio), so that the weight on stocks is the same in every version. CAPE is a real valuation measure, so stocks do not change between the nominal and real versions. For $r^*$ I use the New York Fed's current Laubach–Williams estimates (Holston et al., 2023). These "one-sided" estimates use only data up to each date, with the model's parameters estimated on the full sample.

**The two-year yield.** The two-year yield is adjusted in the same three versions. With $y_t^2$ the nominal yield and $\pi_t^2$ two-year expected inflation (from the Cleveland Fed; Haubrich et al., 2012),

$$R_t^{\text{nominal}} = y_t^2, \qquad R_t^{\text{real}} = y_t^2 - \pi_t^2, \qquad R_t^{r^*\text{-gap}} = y_t^2 - \pi_t^2 - r_t^*. \tag{8}$$

Expected inflation and $r^*$ are monthly or quarterly and move slowly. The daily series add their interpolated monthly values to daily market prices, so day-to-day moves come from markets (Appendix F).

## 2.2 Why financial conditions and $r^*$

Policy reaches spending and prices through many interest rates, asset prices and the dollar. A standard VAR with one interest rate can miss most of that route, and the omission can produce the puzzles.

### 2.2.1 The standard transmission mechanism

Monetary policy sets the short-term interest rate today and, through its communications, shapes what people expect that rate to be in the future. Few households and businesses borrow at the policy rate. They face the rates on consumer credit, mortgages and business debt, which arbitrage links to current and expected short-term rates. The same rise in rates raises the return investors require on stocks. For businesses, a higher cost of capital raises the user cost of new equipment and buildings, and investment falls (Jorgenson, 1963; Hall and Jorgenson, 1967). The value of existing capital relative to its replacement cost falls as well (Tobin, 1969). For households, lower stock prices lower wealth, and consumption depends on wealth as well as on current income (Ando and Modigliani, 1963; Friedman, 1957). Higher rates also strengthen the dollar, which shifts spending toward imports and reduces exports (Fleming, 1962; Mundell, 1963).

Interest rates of many maturities, equity prices and the dollar therefore form the core of the transmission mechanism (Boivin et al., 2010). The same links appear in modern models of the business cycle, in closed and open economies (Christiano et al., 2005; Smets and Wouters, 2007; Galí and Monacelli, 2005). Models with heterogeneous households give more weight to the indirect effects that work through income (Kaplan et al., 2018; Auclert, 2019). But in all of these models

the short-term rate matters mainly because of what it does to the prices and rates that people actually face. The effects come with a lag (Friedman, 1961).

What matters in these models is not the nominal rate but the real rate relative to the neutral rate $r^*$. The same holds for the other parts of financial conditions: a given yield or stock valuation is tighter when $r^*$ is low than when it is high. The distinction matters most for inflation. Spending responds to financial conditions within months. Inflation responds to the slack that builds up over time, so it depends on how long, and by how much, conditions stay tight relative to $r^*$. A measure of conditions that ignores a falling $r^*$ will overstate how tight policy has been, and by more the longer the period. That error is largest for the cumulative effects that drive inflation.

### 2.2.2 What standard VARs leave out

The workhorse evidence on monetary policy comes from VARs with a few macroeconomic variables and one interest rate (Bernanke and Blinder, 1992; Christiano et al., 1999). Some studies add a credit spread (Gertler and Karadi, 2015; Caldara and Herbst, 2019), and factor models bring in many more series (Bernanke et al., 2005). But long-term yields, equity valuations and the dollar are usually absent from the small models used to measure the effects of policy, although some recent work adds the dollar and stock-market volatility (Boehm et al., 2026). These are the variables that, on the mechanism above, carry those effects. Concern about missing variables is not new: Sims (1992) argued that the price puzzle arises because the Federal Reserve responds to information that small VARs do not contain (see also Castelnuovo and Surico, 2010). My concern is different. Here the missing variables are the channel itself.

A simple two-equation model (Appendix A) shows what this omission can do. If unemployment responds to financial conditions, and financial conditions respond to the policy rate and to other forces, a VAR in unemployment and the policy rate lets the rate stand in for financial conditions. Its error contains past financial shocks, which a VAR with few lags cannot capture, so the VAR is misspecified (on missing variables in VARs, see Fernández-Villaverde et al., 2007) and the effects of policy are harder to measure.

A second problem matters more for the size and sign of the response: the omission changes the experiment being traced. In the windows, the FCI used below rises about 1 unit per percentage point rise in the two-year yield. In an ordinary month, an unexpected rise of 1 percentage point in the two-year yield, measured by the residual of a monthly VAR in unemployment, inflation, the yield and the FCI, comes with a rise in the FCI of only 0.36 to 0.51 unit in the real and $r^*$ versions (0.38 in the nominal version). A VAR without financial conditions traces the typical move in the two-year yield, which comes with a much smaller change in financial conditions than a policy surprise does.

High-frequency surprises help identify the shock, but they do not by themselves fix how it is carried forward. My design therefore does two things. It includes financial conditions in the system, so that they carry the effects of the surprise forward, as in equation (14). And it measures the immediate effect of the surprise on the two-year yield and on financial conditions directly in the window, where policy news is the main force at work. The measurement of $f_t$ also matters. If activity responds to financial conditions relative to $r^*$, a measure that ignores $r^*$ differs from the true $f_t$ by a slow-moving error, and that error enters equation (14).

### 2.2.3 Financial conditions and real activity

A separate literature asks how financial conditions are related to economic activity. Asset prices help forecast output and inflation, but which ones help changes over time (Stock and Watson,

2003). Credit spreads, and especially their excess bond premium component, are strong predictors of activity (Gilchrist and Zakrajšek, 2012). Broad indexes summarize many such series and are used to assess the outlook (Hatzius et al., 2010; Brave and Butters, 2011). Financial conditions are also linked to the risks around growth, not only to its average (Adrian et al., 2019), and their relationship with unemployment may be nonlinear (Kiley, 2020a). Federal Reserve staff now publish a measure of the effect of financial conditions on growth, with weights from large macroeconomic models (Ajello et al., 2023).

This work has a limit that the event-study literature does not. A correlation between financial conditions and later activity need not reflect a causal effect, because financial conditions also respond to the economy. Hatzius et al. (2010) address part of this problem by removing the effects of past growth and inflation from their indicators. The companion note uses the timing in a VAR for the same purpose (FCI note, 2026). The two literatures have complementary strengths. Work on high-frequency surprises identifies policy shocks cleanly but carries them through a small set of variables. Work on financial conditions studies a broad set of variables but does not isolate policy shocks. I bring them together: the weights in my index come from how its components respond to policy news, and the surprise is carried forward through the estimated relationship between that index and the economy.

#### 2.2.4 Policy news beyond interest rates, and risk appetite

Recent work finds that announcements move stock prices and the dollar much more than interest-rate surprises can explain (Boehm and Kroner, 2024; Bauer et al., 2026). My windows show the same pattern. The policy-news surprise explains 80 percent of the variance of the move in the two-year yield, but only 32 percent for the 10-year yield, 31 percent for the S&P 500, 34 percent for the dollar and 48 percent for the FCI. The part of the FCI's move that the surprise does not explain is therefore large.

There are two readings of this unexplained part. On one, it is mostly news unrelated to policy that happens to arrive in the window, and my estimates capture the effects of policy. On the other, much of it is policy news that does not move interest rates, as Boehm and Kroner (2024) and Bauer et al. (2026) argue. Then my estimates are the effects of one dimension of policy news, and the effects of all policy news through financial conditions could be larger. Both readings agree on the channel: policy news reaches the economy through broad financial conditions.

**[TO DO: The earlier draft also reported (i) the window ratio by subperiod and by VIX/EBP state, (ii) the effect of a one-standard-deviation move in the unexplained part of the FCI, (iii) Baa/EBP credit, FCI-components and commodity-price robustness. None of these is in the clean package; (iii) credit moves to the EBP robustness subsection (Section 4); regenerate (i) and (ii) in baseline/ only if the paper needs them.]**

### 2.3 Timing assumptions and identification

The shock is the Nakamura–Steinsson policy-news surprise, $s_a$ in equation (1), from the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). I use the 252 FOMC announcements from January 1996 to July 2026 with all the window measures available (Appendix F). The window runs from 10 minutes before the statement to 60 minutes after the press conference starts, or 30 minutes around the statement when there is no press conference. Identification comes from timing, at three frequencies. Table 1 summarizes the assumptions.

**Table 1:** Timing assumptions

| Frequency | What can move | Assumption | Equation |
|---|---|---|---|
| Window (minutes) | Two-year yield and FCI, by the amounts measured in the window | News about the path of interest rates is the main systematic force in the window, and the only one correlated with the surprise. Unemployment and inflation do not move. | (10) |
| Daily | Two-year yield and FCI, each day after the announcement | They depend on their own recent history and on the unemployment and inflation of this month and last month. | (11) |
| Monthly | Unemployment and inflation, from the month after the announcement | Within a month, unemployment and inflation are set first. They respond to financial variables only with a lag of at least one month. | (12) |

**The window.** The window move in the FCI is computed with the weights in equation (3):

$$\Delta F_a = w_B \, \Delta y_a^{10} + w_Q \, \Delta \mathrm{SP}_a + w_D \, \Delta \mathrm{DXY}_a. \tag{9}$$

These are the measures available within the day, and the ones used to estimate the weights. Percent changes in the S&P 500 equal percent changes in CAPE within a window, because earnings are fixed. I express all effects per percentage point on the two-year yield in the window. The effect on the FCI is the ratio of two regression slopes on the surprise, an instrumental-variables estimate with $s_a$ as the instrument:

$$b = \begin{pmatrix} 1 \\ b_F \end{pmatrix}, \qquad b_F = \frac{\sum_a (\Delta F_a - \overline{\Delta F})\,(s_a - \bar{s})}{\sum_a (\Delta y_a^2 - \overline{\Delta y^2})\,(s_a - \bar{s})} = 0.99. \tag{10}$$

This is close to one because the weights are normalized that way. It is not exactly one because this regression includes a constant, which the scaling in equation (1) does not, and because it uses one fewer announcement. Expected inflation and $r^*$ move slowly, so I take the window move to be the same in every version.

**The day.** On the announcement day, the two-year yield and the FCI shift by the vector $b$. I use the window move rather than the change over the whole day, because the full-day change contains other news. On later days, the two series move as equation (11) implies. They affect each other only through the daily lags, so no ordering between them is needed within the day. The window fixes their joint move.

**The month.** Within each month, unemployment and inflation are set first, and markets then respond to them. So in the month of the announcement, unemployment and inflation have not moved. They respond from the next month on, through the monthly average of the daily financial series. That average depends on when in the month the announcement falls: an announcement late in a month affects few days of that month's average. This is how the model takes the timing of announcements into account, as Jacobson et al. (2026) recommend. The ordering is the familiar recursive one, in which slow-moving macroeconomic variables come before financial variables (Bernanke and Blinder, 1992; Christiano et al., 1999). It also means that the daily equations include the current month's unemployment and inflation, even though those figures are published later. I treat this as a modeling convention: the monthly values summarize conditions that markets observe as the month unfolds.

**The key assumptions.** Two assumptions carry the results. The first is the usual one in a VAR. Once the shock has been identified, the economy responds to a change in the financial variables in the same way whatever caused it. A policy shock identified by a Cholesky ordering or an external instrument is carried through the same estimated dynamics as every other shock, and I do the same. This has a familiar cost. Stock prices partly anticipate future activity, so a move that policy news pushes through stocks need not act on the economy like a move with another cause. This is the same problem as the information effect in any VAR with a high-frequency shock; as in the companion note, I set it aside.

The second assumption is specific to this paper and has two parts. A standard VAR estimates the effect of the shock on every variable in the system. Here, the surprise reaches unemployment and prices only through the two-year yield and the FCI, and has no direct effect on them in the window. Anything else it moves, such as credit spreads or expectations, is left out. So are the moves in stock prices and the dollar in the window that are unrelated to $s_a$; the surprise explains only about a third of those moves (Section 2.2.4). The non-yield shock of Boehm and Kroner (2024) is uncorrelated with policy-news surprises, so such moves do not bias $b_F$. But the weights in equation (3) treat them as noise, and part of them may be policy news (Boehm and Kroner, 2024; Bauer et al., 2026). In addition, the monthly equations see the FCI as a single index. Its weights show how the three components respond to policy news, not how the economy responds to each. So I assume that the three affect unemployment and prices in proportion to those weights.

## 2.4 A mixed-frequency VAR

**Structure.** The model has a daily block and a monthly block. Let $x_\tau = (R_\tau, F_\tau)'$ hold the two-year yield and the FCI on day $\tau$, in the same version. Let $m_t = (U_t, \pi_t)'$ hold the two monthly variables. The unemployment gap $U_t$ is the unemployment rate minus the CBO estimate of its natural rate. The CBO estimate does not respond to the surprise, so the response of the gap equals that of the unemployment rate. Inflation is $\pi_t = 1200\,\Delta \ln P_t$, where $P_t$ is the price index for personal consumption expenditures excluding food and energy (core PCE).

The daily block is a VAR that uses averages of the past day, week, month and quarter as regressors (Corsi, 2009):

$$x_\tau = \alpha + \sum_{k\in\{1,5,22,66\}} \Phi_k\, \bar{x}^{(k)}_{\tau-1} + \Gamma_0\, m_{t(\tau)} + \Gamma_1\, m_{t(\tau)-1} + \xi_\tau, \tag{11}$$

where $\bar{x}^{(k)}_{\tau-1}$ is the average of $x$ over the $k$ days through $\tau - 1$, and $t(\tau)$ is the month of day $\tau$. Averages over about a day, a week, a month and a quarter capture persistence with few coefficients. When the FCI is left out, $x_\tau$ is the two-year yield alone.

The monthly block is

$$m_t = c + \sum_{l=1}^{L} A_l\, m_{t-l} + \sum_{l=1}^{L} G_l\, \bar{x}_{t-l} + \zeta_t, \qquad \bar{x}_t = \frac{1}{n_t}\sum_{\tau\in t} x_\tau, \tag{12}$$

with $n_t$ the number of weekdays in month $t$. The model is a mixed-frequency VAR (Ghysels, 2016). The equation contains no current-month term in $\bar{x}_t$, which is the timing assumption of Table 1.

A monthly VAR with the same variables and a recursive ordering is the special case in which every announcement falls on the first day of its month (Appendix B).

**The time trend.** The baseline monthly equations contain a constant and no time trend. Over 1996–2026, unemployment, inflation and interest rates contain slow movements that a short monthly model may not capture: rates drifted down along with $r^*$, inflation was low in the 2010s

and rose after 2020, and the natural rate of unemployment may have shifted in ways the CBO estimate misses.

**Estimation and inference.** I estimate each block by ordinary least squares, on the same sample: January 1996, the month of the first announcement with all window measures, to September 2026 (369 months, and the weekdays of those months for the daily block). The lagged daily averages and the monthly lags may use earlier observations. 2020 is kept in both blocks. The baseline has $L = 12$ monthly lags, fixed in advance (Appendix D shows 3 and 6). Responses are computed recursively, as deviations from a path with no surprise, and averaged over the positions of the announcements within their months (Appendix F); the response of the price level after $h$ months is $\sum_{j=0}^{h} \tilde{\pi}_j/12$. Intervals come from a block bootstrap with 500 draws (Künsch, 1989; Jentsch and Lunsford, 2019) that resamples calendar months in blocks of 12 and uses together the monthly residuals, the daily observations and the announcements of the months drawn (Appendix C). A response is significant at the 5 or 10 percent level when its 95 or 90 percent interval excludes zero. I report 90 percent intervals, as recent studies in this area do (Bauer and Swanson, 2023a; Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021; Ramey, 2016), and 68 percent intervals, which Jarociński and Karadi (2020) report alongside them. The FCI weights and $r^*$ are held fixed.

# 3 Baseline results

## 3.1 Standard methods after the early 1980s

Standard identification schemes have not delivered conventional effects of monetary policy in samples that start after the early 1980s. Ramey (2016) estimates the specification of Christiano et al. (1999) on 1983–2007 and finds that "the results change dramatically and imply that increases in the federal funds rate lower the unemployment rate." A hybrid VAR with the Romer–Romer shock (Romer and Romer, 2004) gives, on the same later sample, contractionary shocks that "appear to be expansionary," as Barakchian and Crowe (2013) found for 1988–2007, and a proxy SVAR with the Romer–Romer shock, estimated on 1969–2007, gives a "pronounced, statistically significant price puzzle" (her Figures 1 and 2). Caldara and Herbst (2019) find, for 1994–2007 with a proxy SVAR that uses high-frequency surprises, that without a credit spread the response of industrial production is weak and uncertain (about $-0.2$ percent, with a credible set that includes positive values), against about $-0.75$ percent when the excess bond premium is added.

I start from the same two standard methods, estimated on my own data and units, so that they can be compared directly with the model below. Both use the monthly unemployment gap, core PCE inflation and the end-of-month two-year Treasury yield, with 12 lags and no trend. The first is the recursive (Cholesky) VAR of Christiano et al. (1999), with the policy rate ordered last, so that unemployment and inflation do not respond to it within the month. The second is a proxy VAR (Stock and Watson, 2018; Mertens and Ravn, 2013; Gertler and Karadi, 2015), which uses the surprise as an instrument for the policy rate: the instrument is the Nakamura–Steinsson policy news (Nakamura and Steinsson, 2018) summed over the announcements of the month, the same 252 announcements used throughout. All responses are to a 1 percentage point rise in the policy rate. Figure 2 and Table 2 show the results.

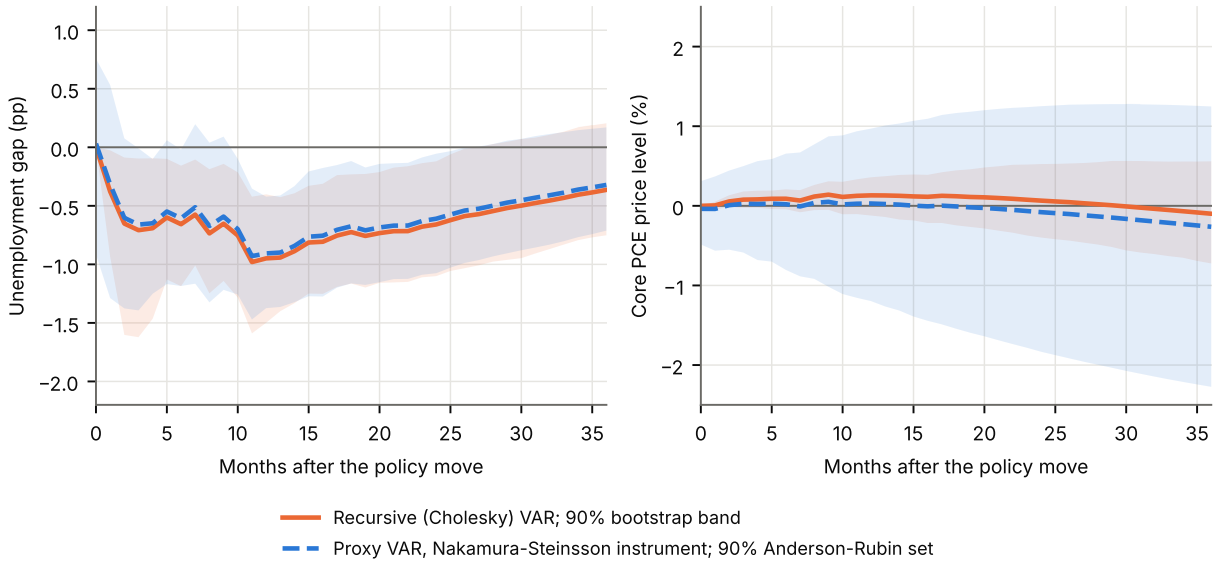



**Figure 2:** Responses to a 1 percentage point rise in the two-year yield, 1996–2026, from a recursive (Cholesky) VAR and a proxy VAR with the Nakamura–Steinsson instrument. The monthly unemployment gap, core PCE inflation and the two-year yield; 12 lags. Bands: 90 percent (bootstrap for the Cholesky VAR, Anderson–Rubin sets for the proxy VAR, truncated at ±4).

**Table 2:** Standard methods after the early 1980s: responses to a 1 percentage point rise in the policy rate

| | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|
| Method | Sample | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Cholesky, 2-year yield | 1996–2026 | 369 | −0.95* | −0.66* | −0.36 | 0.13 | 0.07 | −0.10 |
| | | | [−1.50, −0.40] | [−1.10, −0.12] | [−0.75, 0.21] | [−0.12, 0.36] | [−0.40, 0.52] | [−0.72, 0.56] |
| Cholesky, 2-year yield | 1984–2026 | 513 | −0.61* | −0.36* | −0.16 | 0.11 | 0.15 | 0.13 |
| | | | [−0.99, −0.20] | [−0.59, −0.05] | [−0.37, 0.16] | [−0.06, 0.29] | [−0.16, 0.45] | [−0.30, 0.51] |
| Cholesky, federal funds rate | 1984–2007 | 288 | −0.04 | −0.02 | 0.00 | −0.01 | −0.01 | 0.00 |
| | | | [−0.09, 0.00] | [−0.06, 0.02] | [−0.03, 0.03] | [−0.08, 0.03] | [−0.11, 0.07] | [−0.12, 0.10] |
| Proxy VAR, 2-year yield | 1996–2026 | 369 | −0.91* | −0.61* | −0.32 | 0.03 | −0.08 | −0.26 |
| | | | [−1.37, −0.42] | [−1.08, −0.05] | [−0.71, 0.17] | [−1.20, 0.97] | [−1.83, 1.25] | [−2.27, 1.25] |

Monthly data, 12 lags, constant, no trend; unemployment gap, core PCE inflation and the end-of-month policy rate (the 2-year Treasury yield, or the effective federal funds rate where stated), policy rate ordered last in the Cholesky VARs; price level = cumulated inflation/12. Proxy VAR: Nakamura–Steinsson policy news summed within the month as instrument; first-stage $F = 4.5$. Unit: +1 pp on the policy rate on impact. Brackets: 90% intervals (residual bootstrap for the Cholesky VARs; exact Anderson–Rubin sets for the proxy VAR). * = excludes zero. Specification: baseline/lib/litstyle.py.

In both, unemployment *falls* after a tightening: by 0.95 percentage point after 12 months in the Cholesky VAR and by 0.91 in the proxy VAR. The responses are still negative after 24 months, and the Cholesky VAR gives the same answer when the sample starts in 1984 (−0.61 after 12 months). The price level does not respond in any of them. With the federal funds rate on a 1984–2007 sample, close to the 1983–2007 sample of Ramey (2016), the Cholesky VAR gives the same sign in smaller numbers (−0.04 after 12 months), because a funds-rate innovation is short-lived. Adding a commodity price index, the standard remedy for the price puzzle (Sims, 1992; Ramey, 2016), leaves the two-year-yield response negative (−0.81; Appendix Table A20). The first-stage $F$ statistic of the proxy VAR is 4.5, in the range in which weak-instrument problems matter (Lewis and Mertens, 2026; Bauer et al., 2026), so I report exact Anderson–Rubin sets (Montiel Olea et

al., 2021; Lunsford, 2015), intervals that stay valid when the instrument is weak. These are the responses when the surprises are carried through the two-year yield alone.

### 3.2 The FCI model: baseline results

Figure 3 shows the baseline: the response of the unemployment gap and the core PCE price level to a policy move that raises the two-year yield 1 percentage point in the window, from the model of Section 2.4 with the two-year yield and the FCI both measured relative to $r^*$ (solid line, with 68 and 90 percent bands). The dashed line is the nominal two-year yield alone in the same model, and the dash-dotted line is the Cholesky VAR of Section 3.1. The units are the same: the end-of-month two-year yield moves 0.92 to 1.01 of the window move across the versions, so one percentage point in the window and one percentage point on the end-of-month yield in the monthly VARs are the same shock to within about 8 percent.

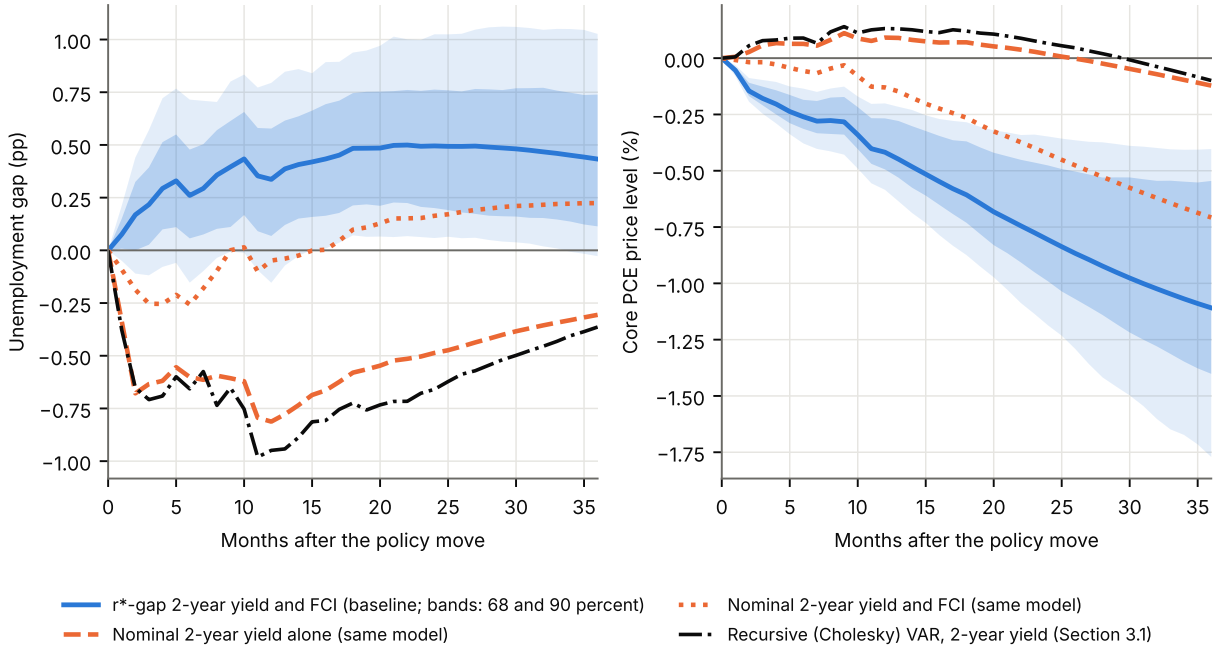



**Figure 3:** Responses to a policy move that raises the two-year yield 1 percentage point: the baseline (two-year yield and FCI relative to $r^*$; 12 lags; shaded bands 68 and 90 percent), the nominal two-year yield alone and with the FCI in the same model, and the Cholesky VAR of Section 3.1.

With the nominal yield alone the model gives what the standard methods give: unemployment falls (to −0.81 percentage point after 12 months) and the price level hardly moves. In the baseline unemployment rises (to 0.50 percentage point after 22 months) and the price level falls steadily (by 1.11 percent after 36 months).

The three versions are reported separately in Table 3 and Figure 4. In each version the two-year yield enters alone and then together with the FCI of the same version: nominal measures; real measures, net of expected inflation; and measures relative to $r^*$, net of expected inflation and of the Laubach–Williams estimate of $r^*$ (Section 2.1 defines them). The tables show 12, 24 and 36 months; the monthly equations have 12 lags, fixed in advance, and Appendix Table A7 shows 3, 6 and 12 lags. The shocks and the monthly equations both start in January 1996, the first announcement for which all window measures are available (369 monthly observations).

**Table 3:** Baseline: 1996 to the present, all data in, no trend (12 lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | none | 12 | 369 | $-0.81^{**}$ | $-0.49^{**}$ | $-0.31$ | 0.09 | 0.02 | $-0.12$ |
| | | | | [−1.22, −0.42] | [−0.80, −0.13] | [−0.65, 0.06] | [−0.09, 0.31] | [−0.33, 0.44] | [−0.62, 0.46] |
| | nominal | 12 | 369 | $-0.05$ | 0.16 | 0.22 | $-0.13$ | $-0.43^{*}$ | $-0.71^{*}$ |
| | | | | [−0.53, 0.37] | [−0.27, 0.54] | [−0.18, 0.62] | [−0.32, 0.08] | [−0.83, −0.02] | [−1.30, −0.07] |
| Real | none | 12 | 369 | $-0.60^{**}$ | $-0.33^{**}$ | $-0.22$ | $-0.03$ | $-0.11$ | $-0.21$ |
| | | | | [−0.94, −0.24] | [−0.63, −0.05] | [−0.51, 0.07] | [−0.18, 0.17] | [−0.36, 0.23] | [−0.58, 0.26] |
| | real | 12 | 369 | 0.09 | 0.23 | 0.21 | $-0.23^{**}$ | $-0.48^{**}$ | $-0.66^{**}$ |
| | | | | [−0.30, 0.46] | [−0.18, 0.60] | [−0.15, 0.62] | [−0.43, −0.06] | [−0.86, −0.12] | [−1.26, −0.10] |
| r*-gap | none | 12 | 369 | $-0.29$ | $-0.07$ | 0.03 | $-0.21^{**}$ | $-0.44^{*}$ | $-0.69^{**}$ |
| | | | | [−0.68, 0.10] | [−0.48, 0.25] | [−0.41, 0.34] | [−0.36, −0.04] | [−0.71, −0.09] | [−1.12, −0.17] |
| | r*-gap | 12 | 369 | 0.34 | $0.50^{*}$ | 0.43 | $-0.42^{**}$ | $-0.81^{**}$ | $-1.11^{**}$ |
| | | | | [−0.15, 0.79] | [0.06, 1.02] | [−0.03, 1.03] | [−0.58, −0.20] | [−1.19, −0.35] | [−1.77, −0.40] |

Nominal, real and r*-gap 2-year yield; the FCI is Note A's index in the matching form in each case. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

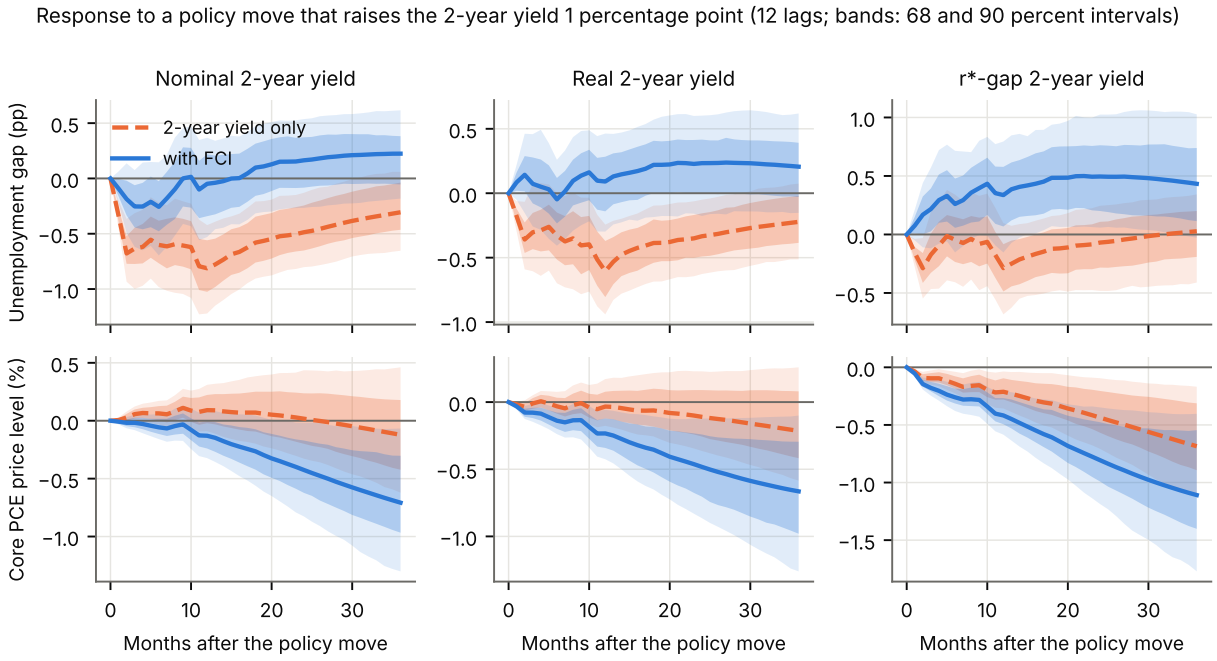



**Figure 4:** Responses to a policy move that raises the two-year yield 1 percentage point, without and with the FCI (12 lags). Bands: 68 and 90 percent intervals.

Adding the FCI removes the fall in unemployment and lowers the price level in every version, and measuring both relative to $r^*$ gives the largest responses. With the nominal or real yield alone, unemployment falls after a tightening (−0.81 and −0.60 percentage point after 12 months) and the price level does not respond. Relative to $r^*$, the yield alone raises no unemployment but lowers the price level (−0.44 percent after 24 months). After 24 months, unemployment with the FCI is 0.50 and the price level −0.81 in the $r^*$ version, against 0.16 and −0.43 in the nominal version and 0.23 and −0.48 in the real version. In the $r^*$ version the unemployment response after 24 months

is significant at the 10 percent level (90 percent interval [0.06, 1.02]) and the price response is significant at the 5 percent level at every horizon.

### 3.3 The implied inflation-unemployment tradeoff

What matters for policy is the ratio of the price response to the unemployment response. Following Barnichon and Mesters (2021), I measure it by the Phillips multiplier at horizon $H$, the cumulative price response per unit of cumulative unemployment response:

$$\mathcal{P}_H = \frac{\tilde{P}_H}{\tilde{U}_H}, \qquad \tilde{U}_H = \frac{1}{12}\sum_{h=1}^{H} \tilde{U}_h, \tag{13}$$

where $\tilde{P}_H$ is the response of the core price level after $H$ months, in percent, and $\tilde{U}_H$ is the cumulative response of the unemployment gap, in percentage-point years. Because the price level cumulates inflation, $\mathcal{P}_H$ is the cumulative change in inflation per unit of cumulative unemployment. Its inverse, $-1/\mathcal{P}_H$, is a sacrifice ratio: the percentage-point years of unemployment needed to lower the price level 1 percent. A more negative $\mathcal{P}_H$ means a steeper Phillips curve.

$\tilde{U}_H$ can be close to zero, and then the ratio is not well defined. So I form 90 percent sets for $\mathcal{P}_H$ in the spirit of Anderson and Rubin (1949): the set contains every value $\mathcal{P}$ for which the 90 percent bootstrap interval of $\tilde{P}_H - \mathcal{P}\,\tilde{U}_H$ contains zero. The sets are computed exactly, not on a grid, and can be unbounded, which shows when the data cannot pin down the tradeoff. Table 4 reports $\mathcal{P}_{24}$ and $\mathcal{P}_{36}$ for each version and for the main sample variations.

**Table 4:** Implied trade-off: Phillips multiplier $\kappa_H$ (12 lags)

| Case | FCI | $H = 24$ months | $H = 36$ months |
|---|---|---|---|
| Nominal | none | −0.01 [−0.29, 0.38] | 0.08 [−0.24, 0.57] |
| | nominal | unbounded | unbounded |
| Real | none | 0.14 [−0.25, 0.82] | 0.21 [−0.19, 1.15] |
| | real | unbounded | unbounded |
| r*-gap | none | unbounded | unbounded |
| | r*-gap | −1.07 [−4.72, −0.49] | −0.90 [−3.71, −0.37] |
| Start 1973 | r*-gap | unbounded | unbounded |
| 2020 dropped | r*-gap | −0.95 [−1.85, −0.38] | −0.82 [−1.94, −0.22] |
| 2020 dropped, start 1973 | r*-gap | −1.49 [−3.37, −0.74] | −1.64 [−5.36, −0.76] |
| Ends 2019 | r*-gap | unbounded | unbounded |
| Ends 2019, start 1973 | r*-gap | −1.21 [−10.65, −0.43] | unbounded |
| r*-gap minus nominal, with FCI | | [−1.55, 7.17] | [−1.50, 5.08] |

$\kappa_H$ = core PCE price level at month $H$ (percent) per percentage-point year of cumulative unemployment-gap response (Barnichon and Mesters 2021); the sacrifice ratio is $-1/\kappa_H$. Point estimate and 90% confidence set (Anderson–Rubin type) in brackets, computed exactly (a wide grid refined by bisection), so a finite end is never an artifact of the grid. An infinite end means that the 90% interval of the cumulative unemployment response includes zero: the data then cannot rule out arbitrarily steep trade-offs on that side; "unbounded": both sides. Last row: paired difference over common announcement draws.

The tradeoff can be measured in the $r^*$ version with the FCI. With the two-year yield and FCI relative to $r^*$, $\mathcal{P}_{24}$ is −1.07 with a 90 percent set of [−4.72, −0.49] and $\mathcal{P}_{36}$ is −0.90 with [−3.71, −0.37]; both sets exclude zero. A sacrifice ratio is then 0.9 percentage-point years of unemployment per 1 percent of price level after 24 months. In the nominal and real versions with the FCI, and for the yield alone relative to $r^*$, the sets are unbounded: the cumulative unemployment response

is too close to zero to pin down the ratio. With the nominal or real yield alone, the multiplier is close to zero and the sets include zero: a tightening that does not raise unemployment says nothing about the cost of lowering inflation. When 2020 is dropped the multiplier is $-0.95$ after 24 months from 1996 and $-1.49$ from 1973, with sets that exclude zero. The set from 1973 ending in 2019 also excludes zero after 24 months, and the sets for the sample from 1973 and for the sample ending in 2019 are unbounded (Table 4). The last row gives the paired difference between the version relative to $r^*$ and the nominal version. For comparison, Barnichon and Mesters (2021) report a multiplier of about $-0.15$ after three to four years for 1990–2008, estimated with high-frequency surprises, and a multiplier that diverges before 1990 and is about $-3$ after five years, estimated with Romer–Romer shocks. Their multiplier is the ratio of the cumulative response of inflation to the cumulative response of unemployment, as in equation (13). At the baseline horizons it lies between the two: $-0.90$ after three years.

### 3.4 Financial conditions in the announcement window

The model assumes that policy reaches unemployment and prices through the two-year yield and the FCI, the broad index of the 10-year yield, equity valuations and the real dollar, weighted by their responses to policy news (Table 5). In the window the FCI moves with the two-year yield: a policy move that raises the two-year yield 1 percentage point raises the FCI by 0.99 (in the same units). A model with the two-year yield alone omits this channel. Figure 5 shows the responses of the two-year yield and the FCI to the policy move in the baseline.

**Table 5:** Market-index weights from policy news

| Component | Response to policy news | Weight | Share of a policy move |
|---|---|---|---|
| 10-year Treasury yield (pp) | +0.53 (0.06) | +0.609 | 0.33 |
| Equity valuation, 100 ln CAPE (%) | −7.78 (0.93) | −0.051 | 0.40 |
| Real broad dollar, 100 ln (%) | +4.22 (0.53) | +0.065 | 0.28 |

Response of each 30-minute announcement-window change to the policy-news factor (scaled so the 2-year yield moves 1 pp), with robust standard errors; weights from the one-factor GLS of Note A; share = weight × response (the three shares sum to 1).

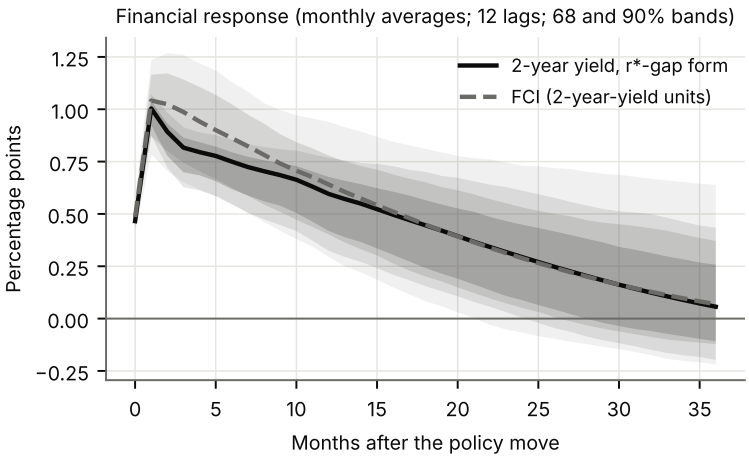



**Figure 5:** Response of the two-year yield and the FCI relative to $r^*$ (in two-year-yield units) to a policy move that raises the two-year yield 1 percentage point in the window (monthly averages; 12 lags; 68 and 90 percent bands).

### 3.5 Estimating the impact from monthly data

A standard proxy VAR can also carry the FCI. It estimates the immediate effect of the surprise on every variable from monthly residuals and the instrument, instead of measuring it in the window. A window-impact VAR takes the effect on the two-year yield and the FCI from the windows, as the main model does, and sets the effect on unemployment and inflation to zero within the month. Table 6 gives the results for the yield and FCI relative to $r^*$; Appendix Tables A22 and A23 give all three versions.

**Table 6:** Monthly VARs with the FCI and the main model ($r^*$-gap two-year yield and FCI; 12 lags)

| | Unemployment gap (pp) | | Price level (%) | |
|---|---|---|---|---|
| | 12 m | 24 m | 12 m | 24 m |
| Proxy VAR | 0.95* | 0.73 | −0.84* | −1.24* |
| | [0.01, 5.87] | [−0.04, 4.55] | [−3.17, −0.15] | [−4.39, −0.24] |
| Window-impact VAR | 0.33 | 0.31 | −0.45* | −0.77* |
| | [−0.25, 0.82] | [−0.25, 0.76] | [−0.71, −0.16] | [−1.26, −0.20] |
| Main model (daily block) | 0.34 | 0.50* | −0.42* | −0.81* |
| | [−0.15, 0.79] | [0.06, 1.02] | [−0.58, −0.20] | [−1.19, −0.35] |

Per +1 pp on the two-year yield. *: the 90% interval excludes zero; intervals in brackets (Anderson–Rubin sets for the proxy VAR; bootstrap for the others). Monthly VARs use end-of-month values.

The proxy VAR has the signs of the main model: unemployment rises 0.95 after 12 months and the price level falls 1.24 percent after 24. Its sets are wide, because the surprise is a weak instrument for monthly residuals (first-stage $F$ of 4.4). The window-impact VAR is close to the main model. To see how much noise to expect, I generate 1,000 samples of 369 months from the estimated baseline model, with the data's calendar and announcement days, and estimate both monthly methods on each (Appendix Table A24; design in Appendix F). The true unemployment response after 12 months is 0.33. The proxy VAR with the FCI is centred on it (median 0.35) with a 5th to 95th percentile range of [−0.31, 1.26], and its first-stage $F$ has a median of 11.1, against 4.4 in the data. Without the FCI the proxy VAR is biased toward zero (median 0.02). The

window-impact VAR is centred on the true response (median 0.35) with a range of [−0.10, 0.80]. Measuring the impact in the window gives the same answer with less noise.

# 4 Robustness

A result about the transmission of monetary policy can rest on one sample, one estimate of $r^*$, one choice of trend, one set of omitted variables or one surprise series. I vary each in turn. Each variation changes one feature of the baseline and keeps the rest: the shocks are the same announcements (except where the surprise itself changes), the monthly equations have 12 lags, and responses are to a policy move that raises the two-year yield 1 percentage point in the window. Table 7 shows the baseline and the 15 variations. Thirteen of them keep the baseline yield and FCI relative to $r^*$; the other two replace the FCI with its three components.

**Table 7:** The baseline and the robustness variations ($r^*$-gap two-year yield; 12 lags)

| | Yield alone | Yield and FCI | |
|---|---|---|---|
| | Unemployment, 12 m | Unemployment, 24 m | Price level, 24 m |
| Baseline | −0.29 | 0.50* | −0.81** |
| *Sample* | | | |
| Start 1973 | −0.05 | 0.14 | −0.63** |
| 2020 dropped | −0.10 | 0.45** | −0.69** |
| 2020 dropped, start 1973 | 0.17** | 0.24 | −0.76** |
| Ends 2019 | −0.15 | 0.13 | −0.14 |
| Ends 2019, start 1973 | 0.19** | 0.16 | −0.50** |
| *The $r^*$ adjustment and trends* | | | |
| Kiley $r^*$ | −0.53** | 0.28 | −0.18 |
| Kiley $r^*$, start 1973 | −0.11 | −0.04 | −0.37** |
| Real yield and FCI, spline trend | −0.64** | 0.28 | −0.25 |
| Real yield and FCI, spline trend, start 1973 | −0.10 | 0.18 | −0.39** |
| *Credit conditions* | | | |
| Excess bond premium added | −0.08 | 0.52* | −0.78** |
| Excess bond premium added, start 1973 | −0.03 | 0.03 | −0.57** |
| *Weights of the FCI* | | | |
| Three components in place of the FCI | – | 0.22 | 0.06 |
| Three components, start 1973 | – | 0.16 | −0.60 |
| *The surprise* | | | |
| Bauer–Swanson surprise | −0.29 | 0.54** | −0.84** |
| Jarocinski–Karadi surprise | −0.29 | 0.56** | −0.85** |

Per +1 pp on the two-year yield; unemployment gap in percentage points, price level in percent. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero). The variations are described in Sections 4.1 to 4.5; Appendix Tables give each at 12, 24 and 36 months and at 3, 6 and 12 lags. The three-component rows replace the FCI, so they have no yield-alone entry.

Across the 13 variations that keep the FCI relative to $r^*$, the price level falls in every one after 24 months and the fall is significant at the 5 percent level in 10. Unemployment rises in 12 of the 13 and is significantly higher at the 5 or 10 percent level in 4, and no variation gives a significant fall. Without the FCI, unemployment after 12 months falls in 11 of the 13, the wrong-signed response that the FCI removes. The price response is not significant in three variations, each from 1996: the

sample that ends in 2019, Kiley's $r^*$ and the spline trend with real measures. The three-component variation from 1996 is the other case in which the price result does not carry over. The subsections say what each variation tests and what it shows.

### 4.1 The sample

Three features of the sample could drive the result. It is short, so the monetary environment since 1996 may be special. It contains the pandemic, when unemployment moved by more than in any other period. And it ends with 2022–24, the years of the largest tightening in the sample, so the result could rest on them. In the first variation the monthly equations start in 1973 rather than 1996; the months before 1990, for which there are no daily data, use monthly series in place of the daily block, the daily block is still estimated from 1996, and the shocks are still the announcements from 1996. In the second, March to December 2020 are dropped from the monthly and daily equations (2020 values still enter as lags). In the third, all equations end in December 2019. Each is run with the monthly equations starting in 1996 and in 1973.

Starting in 1973 leaves the price response as in the baseline ($-0.63$ after 24 months) and unemployment close to zero. Dropping 2020 leaves the price response ($-0.69$) and makes the unemployment response larger (0.45, significant at the 5 percent level). Ending the sample in 2019 is the one test the price response does not pass from 1996: it is $-0.14$ after 24 months, with a 90 percent interval of $[-0.41, 0.09]$, so from 1996 the fall comes from the years after 2019. From 1973 the same end date gives $-0.50$, significant at the 5 percent level. When the monthly equations start in 1973 the yield alone moves unemployment little or raises it, so the wrong-signed response that the FCI removes belongs to the sample since 1996. Appendix Table A2 gives the full results.

### 4.2 The $r^*$ adjustment and trends

The neutral rate is not observed, and the baseline uses one estimate of it, that of Laubach and Williams. Two variations check that the result does not depend on that choice. The first replaces it with Kiley's estimate, in the two-year yield and in the FCI alike. The second uses no $r^*$ measure at all: it keeps the real two-year yield and the real FCI and adds a natural cubic spline in time to every monthly equation, with knots at the sample ends and at the NBER business-cycle peaks inside the sample. If the result came from the $r^*$ estimate, it would disappear here.

The FCI removes the fall in unemployment in both. With Kiley's $r^*$ unemployment after 12 months is $-0.53$ without the FCI and 0.39 with it; with real measures and the spline it is $-0.64$ and 0.19. The price response is smaller than in the baseline and, from 1996, not significant ($-0.18$ and $-0.25$ after 24 months). From 1973 it is significant at the 5 percent level in both ($-0.37$ and $-0.39$). The choice of $r^*$ therefore matters for the size of the price response, and the sign pattern of unemployment does not depend on it. Appendix Table A3 gives the full results.

### 4.3 Credit conditions

Credit is the most important channel that the baseline leaves out (Gilchrist and Zakrajšek, 2012; Caldara and Herbst, 2019), and a premium that responds to policy could carry part of what the FCI carries. The variation adds the monthly excess bond premium as a third monthly variable, treated like unemployment and inflation, so that the premium responds to the financial variables and they enter its equation as well.

With the FCI relative to $r^*$ the results are those of the baseline: unemployment rises (0.52 after 24 months) and the price level falls ($-0.78$); from 1973 the price level falls by 0.57 percent. Without the FCI the unemployment response moves toward zero ($-0.08$ after 12 months, against

−0.29 in the baseline), in the direction that adding the FCI moves it. Appendix Table A4 gives the full results.

### 4.4 The weights of the FCI

The baseline assumes that the three parts of the FCI, the 10-year yield, equity valuation and the dollar, affect unemployment and prices in proportion to the weights with which they respond to policy news (Section 2.3). That is an assumption about the economy, not about the surprise, and it can be tested. The variation enters the three weighted components separately in the daily and monthly equations, in the same version as the two-year yield, in place of the FCI, and each has its own window impact; the three impacts add up to the impact of the FCI. To keep the number of parameters reasonable, the daily block is restricted: each component's equation contains its own 1-, 5-, 22- and 66-day averages and those of the two-year yield, plus the monthly macro variables, and the two-year yield's equation its own averages and the macro variables. That is 48 coefficients in all, against 26 in the baseline daily block with the FCI. The monthly equations have 12 lags and start in 1996 or, as a further variation, in 1973. A Wald test of the restriction that the coefficients on the three components are equal at each lag, with bootstrap $p$-values, tests the assumption directly (Table 8).

In the $r^*$ version unemployment still rises (0.20 after 12 months from 1996 and 0.17 from 1973), with wider intervals because the variation adds many coefficients. The price response differs by sample. From 1973 it is −0.35 after 12 months, significant at the 5 percent level, as with the FCI. From 1996 it is close to zero after 24 months (0.06, with a 90 percent interval of [−0.25, 0.40]). This is the one variation from 1996 in which the price result does not carry over. The test agrees. The restriction is not rejected in the unemployment equation. In the inflation equation it is rejected from 1996 (bootstrap $p$ of 0.013 in the $r^*$ version) and not from 1973 (0.374). From 1996 the price result therefore depends on the assumption of proportional weights, and from 1973 it does not. Appendix Table A5 gives the full results.

**Table 8:** Test of the proportional-weights restriction (12 lags)

| | Unemployment gap equation | | Core PCE inflation equation | |
|---|---|---|---|---|
| Step | Wald | Bootstrap $p$ | Wald | Bootstrap $p$ |
| Nominal | 8.9 | 0.984 | 41.9 | 0.053 |
| Real | 7.0 | 0.996 | 42.9 | 0.041 |
| r*-gap | 21.7 | 0.399 | 53.9 | 0.013 |
| Nominal, start 1973 | 24.5 | 0.930 | 26.0 | 0.539 |
| Real, start 1973 | 22.2 | 0.886 | 32.4 | 0.232 |
| r*-gap, start 1973 | 33.4 | 0.408 | 27.9 | 0.374 |

Wald statistic (heteroskedasticity-robust, HC1) for the restriction that the coefficients on the three weighted components (10-year yield, equity valuation, dollar) are equal at each lag (24 restrictions), in each monthly equation, and the bootstrap $p$-value (999 residual draws under the restriction); monthly equations from 1996 (first three rows) and from 1973 (last three), no trend, 2020 in. The unrestricted equation is the one used in Robustness 4.

### 4.5 The surprise

The baseline uses the Nakamura–Steinsson policy news. An announcement can reveal the Fed's view of the economy as well as a change in policy, and part of each surprise can be predicted from public data. Two variations change the surprise. The first uses the orthogonalized surprise of Bauer and Swanson (2023a), which removes the part of each surprise that public information

predicts; it is available for announcements through 2019 (201 announcements). The second keeps the Nakamura–Steinsson surprise only for announcements in which the two-year yield and the S&P 500 move in opposite directions in the window, the sign restriction of Jarociński and Karadi (2020) (157 announcements).

With the two-year yield alone the responses are the same as in the baseline, because the shock is scaled to the yield. With the FCI they are larger: unemployment rises 0.54 and 0.56 after 24 months, against 0.50 in the baseline, and the price level falls 0.84 and 0.85 percent, against 0.81. Both are significant at the 5 percent level. Appendix Table A6 gives the full results.

# 5 Conclusion

Interest-rate surprises around Federal Reserve announcements measure policy shocks well, at least the part of policy news that moves interest rates. The hard part has been tracing their effects on the economy. I kept the surprise as it is and changed how its effects are traced: through broad financial conditions rather than one interest rate, and relative to the neutral rate $r^*$. Traced this way, the responses of unemployment and prices have the standard signs, where the nominal two-year yield alone and the standard methods give the wrong ones. The ratio of the price response to the unemployment response is of conventional size: lowering the core price level 1 percent costs about 0.9 percentage-point years of unemployment over two years.

Earlier work improves the surprise, by removing the parts that public information predicts or that reveal the Fed's view of the economy (Bauer and Swanson, 2023a; Jarociński and Karadi, 2020), or improves the timing of the outcome (Jacobson et al., 2026), and recent work adds a second, non-yield shock that moves stock prices and the dollar (Bauer et al., 2026; Boehm and Kroner, 2024; Boehm et al., 2026). I leave the surprise as it is and change the channel. The risk-appetite work and mine agree that stock prices and the dollar carry much of the effect of policy; they differ in whether those moves are a separate shock or part of financial conditions. Measuring conditions relative to $r^*$ gives the largest responses and the tradeoff.

For policy, the results support judging the stance by financial conditions relative to $r^*$, as the FOMC did in 2023 (Powell, 2023), and they explain why the same policy rate can be more or less restrictive at different times. In round numbers, a policy move that raises the two-year yield 1 percentage point lowers the core price level by 0.81 percent after two years. The estimates are effects of surprises, not a decomposition of any episode.

The broader lesson is that the fall in $r^*$ over the past three decades is more than a fact about the level of interest rates. Unless rates are measured relative to $r^*$, that fall also makes the effects of policy, and the cost of lowering inflation, harder to measure.

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**[TO DO: Appendix: the 3/6/12-lag tables now come from baseline/. Still to check: the data-sources table and the construction text against baseline/data/SOURCES.md (unverified items: EBP file origin, r* vintage, daily CAPE and dollar file, DFF and PPIACO copies).]**

# A A two-equation illustration

Let $U_t$ be the unemployment gap, $i_t$ the policy rate and $f_t$ financial conditions, with a higher $f_t$ meaning tighter conditions:

$$U_t = \rho_U \, U_{t-1} + \kappa \, f_{t-1} + \varepsilon_t^U, \tag{14}$$

$$f_t = \rho_f \, f_{t-1} + \lambda \, i_t + \varepsilon_t^f. \tag{15}$$

Unemployment responds to financial conditions and not directly to the policy rate. Financial conditions respond to the policy rate and to other forces, $\varepsilon_t^f$, such as changes in risk appetite. Some of these other forces may themselves be policy news that does not move interest rates (Boehm and

Kroner, 2024; Bauer et al., 2026). Suppose the economist observes only $U_t$ and $i_t$. To remove $f_t$, lag equation (14) one month, multiply it by $\rho_f$, subtract it from equation (14), and use equation (15). This gives

$$U_t = (\rho_U + \rho_f)\, U_{t-1} - \rho_U \rho_f\, U_{t-2} + \kappa\lambda\, i_{t-1} + \underbrace{\varepsilon^U_t - \rho_f\, \varepsilon^U_{t-1} + \kappa\, \varepsilon^f_{t-1}}_{\text{error}}. \tag{16}$$

The policy rate has no direct effect in this model. It appears in equation (16) because it stands in for financial conditions. The error is a moving average that includes past financial shocks, so a VAR in $U_t$ and $i_t$ with few lags cannot capture it. The VAR is misspecified. With the lags and samples available in practice, this can make the effects of monetary policy harder to measure.

# B Relation to a monthly VAR and a proxy VAR

Suppose the daily block were replaced by a monthly equation for $\bar{x}_t$, and every announcement occurred on the first day of the month. With $Z_t = (m_t', \bar{x}_t')'$, the model would then be a standard monthly VAR with a recursive ordering, the macroeconomic variables first, and impact vector $(0, 0, b')'$. The daily block adds two things: the position of each announcement in its month, and fast dynamics of the financial variables, estimated on many more observations. A proxy VAR (Gertler and Karadi, 2015; Mertens and Ravn, 2013; Stock and Watson, 2018) instead estimates $Z_t = \sum_l C_l Z_{t-l} + \iota_t$ and uses as an instrument $z_t$ the surprises summed over the month. The effect on the FCI per unit effect on the two-year yield is then

$$\frac{\mathrm{Cov}(\iota_{F,t}, z_t)}{\mathrm{Cov}(\iota_{R,t}, z_t)}. \tag{17}$$

The window ratio $b_F$ in equation (10) has the same form, with three differences. It replaces monthly VAR residuals with changes measured inside the window, which do not depend on how strongly the surprise predicts monthly residuals. It sets the entries for unemployment and inflation to zero rather than estimating them. And it dates each shock to its day rather than summing shocks over a month. Section 3.5 estimates both monthly benchmarks. In large samples a VAR of this kind estimates the same effects as local projections (Plagborg-Møller and Wolf, 2021). In the language of mixed-data sampling, the monthly equation gives equal weights to the days within a month and unrestricted weights to the months (Ghysels et al., 2007; Foroni et al., 2015).

# C The bootstrap

Every interval in the note comes from a block bootstrap. Each draw resamples the 369 calendar months from January 1996 in moving blocks of 12 months and uses together the monthly residuals, the daily observations and the announcements of the months drawn, which keeps the dependence between them. The announcements drawn give new values of $b$ and new positions. The monthly equations are re-estimated on a series rebuilt recursively from the drawn residuals, with the macro lags taken from the simulated series and the financial lags from the data, and the daily block is re-estimated on the drawn daily observations. When the monthly equations start in 1973, the residuals before 1996 are drawn in blocks of 12 months of their own. A month without a monthly observation (2020 dropped, or a sample ending in 2019) contributes its daily observations and announcements but no residual. A simpler bootstrap resamples the three sources of sampling variation separately: the announcements and the monthly residuals independently across months (with the monthly lags held at their values in the data), and the daily observations in moving blocks of 66 days. I ran it

with 12 lags and 500 draws for the nominal, real and $r^*$-relative two-year yield, each without and with the FCI of the same version. Across the 36 responses at 12, 24 and 36 months, its 90 percent intervals are wider than those of the block bootstrap in 34, by a median of 1.15 times (range 0.99 to 1.40). The two agree on whether a response is significant at the 10 percent level in 34 of the 36 responses and at the 5 percent level in 30. The price response of the baseline is significant at the 5 percent level at every horizon under both. The 90 percent interval of the baseline unemployment response after 24 months is [0.06, 1.02] under the block bootstrap and [0.00, 1.02] under the simpler one.

# D Lag length

The baseline uses 12 monthly lags, fixed in advance. Table A1 repeats the main results with 3 and 6 lags. The sample, the announcements and the bootstrap are the same at every lag length: the block bootstrap with 500 draws. The tables that follow give every case at 3, 6 and 12 lags.

The direction of every result is the same at every lag length. With the FCI, the price level falls in all 48 responses (16 cases at 12, 24 and 36 months), and no response of unemployment is significantly negative. Unemployment rises in 34 of the 48 responses with 3 lags, 43 with 6 and 42 with 12. Without the FCI, the Cholesky and proxy VARs show a fall in unemployment at every lag length; the Cholesky fall at 12 months is significant at all three. The weights test gives the same verdicts: in the sample from 1996 it rejects proportional weights in the inflation equation at the 10 percent level and does not reject them in the unemployment equation at the 5 percent level, and in the sample from 1973 it rejects in neither. In the simulation, the median proxy-VAR estimate of the unemployment response with the FCI is close to the true response at every lag length.

The size of the responses and the strength of the evidence increase with the lag length. The median price response with the FCI across the 48 responses is $-0.21$ with 3 lags, $-0.35$ with 6 and $-0.44$ with 12. The baseline price response at 36 months is $-0.27$, $-0.87$ and $-1.11$ percent. The number of significant price responses is 4, 30 and 39 of 48. The Phillips multiplier is bounded at 12 lags only; with 3 or 6 lags the data do not rule out arbitrarily steep trade-offs.

**Table A1:** The main results with 3, 6 and 12 monthly lags

| | 3 lags | 6 lags | 12 lags |
|---|---|---|---|
| *Standard methods (per +1 pp on the policy rate)* | | | |
| Cholesky, 2-year yield: unemployment gap, 12 m | −0.64* [−1.01, −0.15] | −0.56* [−0.97, −0.12] | −0.95* [−1.50, −0.40] |
| Cholesky, 2-year yield: price level, 36 m | 0.09 [−0.49, 0.47] | −0.08 [−0.74, 0.36] | −0.10 [−0.72, 0.56] |
| Proxy VAR: unemployment gap, 24 m | −0.43* [−0.64, −0.10] | −0.36 [−0.76, 0.34] | −0.61* [−1.08, −0.05] |
| *FCI model, r*-gap 2-year yield (baseline)* | | | |
| With FCI: unemployment gap, 24 m | 0.13 [−0.27, 0.48] | 0.36 [−0.09, 0.79] | 0.50* [0.06, 1.02] |
| With FCI: price level, 24 m | −0.25 [−0.64, 0.17] | −0.63* [−1.04, −0.15] | −0.81* [−1.19, −0.35] |
| With FCI: price level, 36 m | −0.27 [−0.78, 0.26] | −0.87* [−1.49, −0.18] | −1.11* [−1.77, −0.40] |
| Without FCI: unemployment gap, 12 m | −0.17 [−0.47, 0.11] | −0.09 [−0.41, 0.22] | −0.29 [−0.68, 0.10] |
| Phillips multiplier, 24 m | −0.65 [unbounded] | −1.07 [unbounded] | −1.07 [−4.72, −0.49] |
| *All cases with the FCI in Sections 3 and 4 (16 cases, 3 horizons, 48 responses each)* | | | |
| Price level falls | 48 of 48 | 48 of 48 | 48 of 48 |
| Price level falls significantly | 4 of 48 | 30 of 48 | 39 of 48 |
| Unemployment gap rises | 34 of 48 | 43 of 48 | 42 of 48 |
| Unemployment gap rises significantly | 3 of 48 | 9 of 48 | 10 of 48 |
| *Financial weights and simulation (r*-gap)* | | | |
| Proportional weights, 1996: bootstrap $p$, unemployment equation | 0.201 | 0.236 | 0.399 |
| Proportional weights, 1996: bootstrap $p$, inflation equation | 0.017 | 0.050 | 0.013 |
| Proportional weights, 1973: bootstrap $p$, inflation equation | 0.609 | 0.752 | 0.374 |
| Proxy VAR with FCI, simulated: median unemployment, 12 m (true value) | 0.25 (0.25) | 0.36 (0.33) | 0.35 (0.33) |
| Proxy VAR with FCI, simulated: first-stage $F$, median | 12.2 | 11.8 | 11.1 |

Same sample (1996 to 2026), announcements and block bootstrap (500 draws) at every lag length. Responses per +1 pp on the policy rate; **, *: 95% and 90% intervals exclude zero; 90% intervals in brackets. Counts (at the 10 percent level): the 16 cases with the FCI in the baseline and Robustness 1, 2, 3 and 5 (components excluded), responses at 12, 24 and 36 months.

# E Additional tables

**Table A2:** Robustness 1: sample (r*-gap 2-year yield and FCI) (12 lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Start 1973 | none | 12 | 633 | −0.05 | −0.01 | 0.00 | −0.22** | −0.50** | −0.79** |
| | | | | [−0.21, 0.09] | [−0.16, 0.13] | [−0.16, 0.13] | [−0.32, −0.09] | [−0.70, −0.17] | [−1.11, −0.26] |
| | r*-gap | 12 | 633 | 0.18 | 0.14 | 0.13 | −0.27** | −0.63** | −0.98** |
| | | | | [−0.08, 0.38] | [−0.15, 0.35] | [−0.17, 0.35] | [−0.38, −0.10] | [−0.83, −0.24] | [−1.32, −0.30] |
| 2020 dropped | none | 12 | 359 | −0.10 | −0.21 | −0.25 | −0.20** | −0.32* | −0.44 |
| | | | | [−0.28, 0.08] | [−0.48, 0.08] | [−0.59, 0.12] | [−0.33, −0.04] | [−0.60, −0.01] | [−0.89, 0.03] |
| | r*-gap | 12 | 359 | 0.41** | 0.45** | 0.40 | −0.38** | −0.69** | −0.94** |
| | | | | [0.20, 0.58] | [0.08, 0.73] | [−0.02, 0.79] | [−0.52, −0.17] | [−1.04, −0.24] | [−1.49, −0.20] |
| 2020 dropped, start 1973 | none | 12 | 623 | 0.17** | 0.15 | 0.05 | −0.28** | −0.64** | −0.97** |
| | | | | [0.05, 0.29] | [−0.02, 0.29] | [−0.13, 0.22] | [−0.37, −0.15] | [−0.82, −0.32] | [−1.31, −0.45] |
| | r*-gap | 12 | 623 | 0.33** | 0.24 | 0.15 | −0.32** | −0.76** | −1.14** |
| | | | | [0.14, 0.46] | [−0.02, 0.45] | [−0.11, 0.39] | [−0.44, −0.16] | [−1.00, −0.38] | [−1.57, −0.50] |
| Ends 2019 | none | 12 | 288 | −0.15 | −0.24 | −0.26 | 0.08 | 0.06 | 0.03 |
| | | | | [−0.38, 0.11] | [−0.58, 0.15] | [−0.68, 0.21] | [−0.05, 0.20] | [−0.12, 0.20] | [−0.19, 0.22] |
| | r*-gap | 12 | 288 | 0.30* | 0.13 | 0.00 | −0.08 | −0.14 | −0.21 |
| | | | | [0.02, 0.49] | [−0.24, 0.48] | [−0.45, 0.44] | [−0.24, 0.06] | [−0.41, 0.09] | [−0.55, 0.08] |
| Ends 2019, start 1973 | none | 12 | 552 | 0.19** | 0.18 | 0.08 | −0.21** | −0.49** | −0.82** |
| | | | | [0.04, 0.30] | [−0.03, 0.32] | [−0.15, 0.24] | [−0.32, −0.06] | [−0.71, −0.15] | [−1.18, −0.25] |
| | r*-gap | 12 | 552 | 0.27** | 0.16 | 0.09 | −0.22** | −0.50** | −0.84** |
| | | | | [0.04, 0.40] | [−0.16, 0.34] | [−0.26, 0.29] | [−0.32, −0.05] | [−0.72, −0.18] | [−1.23, −0.29] |

Start 1973: monthly equations from 1973, with monthly series in place of daily data before 1990 (lib/model.py). 2020 dropped: March to December 2020 removed from the monthly and daily equations (2020 values still enter as lags). Ends 2019: all equations, monthly and daily. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A3:** Robustness 2: the r* adjustment and trends (12 lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | |
| Kiley r* | none | 12 | 369 | −0.53** | −0.32* | −0.23 | −0.09 | −0.10 | |
| | | | | [−0.89, −0.19] | [−0.64, −0.03] | [−0.58, 0.06] | [−0.24, 0.07] | [−0.36, 0.23] | [−0.55 |
| | r*-gap (Kiley) | 12 | 369 | 0.39 | 0.28 | −0.01 | −0.25** | −0.18 | |
| | | | | [0.00, 0.80] | [−0.18, 0.64] | [−0.48, 0.36] | [−0.42, −0.04] | [−0.54, 0.23] | [−0.55 |
| Kiley r*, start 1973 | none | 12 | 633 | −0.11 | −0.05 | −0.03 | −0.18** | −0.35* | − |
| | | | | [−0.25, 0.04] | [−0.18, 0.08] | [−0.16, 0.09] | [−0.28, −0.03] | [−0.57, −0.05] | [−0.87, |
| | r*-gap (Kiley) | 12 | 633 | 0.00 | −0.04 | −0.03 | −0.20** | −0.37** | −0 |
| | | | | [−0.20, 0.14] | [−0.24, 0.11] | [−0.21, 0.10] | [−0.29, −0.05] | [−0.55, −0.10] | [−0.81, |
| Real, spline trend | none | 12 | 369 | −0.64** | −0.34* | −0.20 | 0.03 | −0.06 | |
| | | | | [−1.04, −0.32] | [−0.68, −0.04] | [−0.52, 0.08] | [−0.11, 0.20] | [−0.33, 0.31] | [−0.61 |
| | real | 12 | 369 | 0.19 | 0.28 | 0.17 | −0.14 | −0.25 | |
| | | | | [−0.23, 0.54] | [−0.17, 0.63] | [−0.23, 0.53] | [−0.30, 0.06] | [−0.61, 0.12] | [−0.82 |
| Real, spline trend, start 1973 | none | 12 | 633 | −0.10 | 0.08 | 0.11 | −0.09 | −0.25** | −0 |
| | | | | [−0.27, 0.06] | [−0.14, 0.27] | [−0.11, 0.24] | [−0.16, 0.01] | [−0.40, −0.05] | [−0.60, |
| | real | 12 | 633 | 0.18 | 0.18 | 0.11 | −0.17** | −0.39** | −0 |
| | | | | [−0.04, 0.38] | [−0.11, 0.44] | [−0.18, 0.34] | [−0.25, −0.04] | [−0.59, −0.14] | [−0.82, |

Kiley r*: 2-year minus expected inflation minus Kiley's r*, with Note A's index built with the same r*. Real, spline trend: real 2-year and real FCI with a natural cubic spline trend (knots at the sample ends and at NBER business-cycle peaks inside the sample) in every monthly equation, in place of any r* measure. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A4:** Robustness 3: excess bond premium (r*-gap) (12 lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| EBP added | none | 12 | 369 | −0.08 | 0.08 | 0.16 | −0.27** | −0.53** | −0.78** |
| | | | | [−0.41, 0.22] | [−0.26, 0.37] | [−0.18, 0.45] | [−0.41, −0.09] | [−0.81, −0.18] | [−1.19, −0.27] |
| | r*-gap | 12 | 369 | 0.27 | 0.52* | 0.42 | −0.39** | −0.78** | −1.10** |
| | | | | [−0.18, 0.63] | [0.02, 0.96] | [−0.07, 0.96] | [−0.55, −0.18] | [−1.15, −0.32] | [−1.75, −0.37] |
| EBP added, start 1973 | none | 12 | 633 | −0.03 | −0.01 | 0.01 | −0.22** | −0.50** | −0.78** |
| | | | | [−0.18, 0.08] | [−0.16, 0.09] | [−0.14, 0.13] | [−0.31, −0.09] | [−0.69, −0.16] | [−1.13, −0.23] |
| | r*-gap | 12 | 633 | 0.06 | 0.03 | 0.04 | −0.24** | −0.57** | −0.89** |
| | | | | [−0.13, 0.23] | [−0.19, 0.22] | [−0.17, 0.22] | [−0.34, −0.08] | [−0.76, −0.18] | [−1.21, −0.23] |

The excess bond premium is a third monthly variable, treated like unemployment and inflation (u, inflation, EBP system); responses shown for u and the price level. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A5:** Robustness 4: the FCI replaced by its three weighted components (12 lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | components | 12 | 369 | 0.03 | 0.25 | 0.09 | 0.03 | 0.21 | 0.42 |
| | | | | [−0.55, 0.47] | [−0.35, 0.78] | [−0.50, 0.74] | [−0.17, 0.26] | [−0.21, 0.65] | [−0.26, 1.02] |
| Real | components | 12 | 369 | 0.17 | 0.29 | −0.03 | −0.06 | 0.14 | 0.41 |
| | | | | [−0.31, 0.56] | [−0.22, 0.69] | [−0.51, 0.50] | [−0.24, 0.15] | [−0.21, 0.52] | [−0.17, 0.86] |
| r*-gap | components | 12 | 369 | 0.20 | 0.22 | −0.09 | −0.14 | 0.06 | 0.35 |
| | | | | [−0.30, 0.60] | [−0.29, 0.70] | [−0.69, 0.43] | [−0.28, 0.04] | [−0.25, 0.40] | [−0.22, 0.95] |
| Nominal, start 1973 | components | 12 | 633 | −0.28 | −0.24 | −0.26 | 0.10 | 0.45* | 0.92** |
| | | | | [−0.71, 0.07] | [−0.72, 0.13] | [−0.81, 0.19] | [−0.11, 0.38] | [0.05, 1.21] | [0.22, 2.27] |
| Real, start 1973 | components | 12 | 633 | −0.11 | −0.14 | −0.15 | −0.14 | −0.05 | 0.16 |
| | | | | [−0.57, 0.19] | [−0.61, 0.13] | [−0.67, 0.12] | [−0.30, 0.14] | [−0.37, 0.68] | [−0.36, 1.41] |
| r*-gap, start 1973 | components | 12 | 633 | 0.17 | 0.16 | 0.09 | −0.35** | −0.60 | −0.80 |
| | | | | [−0.33, 0.50] | [−0.38, 0.46] | [−0.37, 0.34] | [−0.51, −0.09] | [−0.89, 0.08] | [−1.24, 0.42] |

Components: the 10-year yield, equity valuation and real dollar, each multiplied by its weight, enter separately in the daily and monthly equations in the same version as the 2-year yield, each with its own window impact (the three impacts sum to the FCI's). Baseline rows without the FCI are in the baseline table. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A6:** Robustness 5: alternative surprises (r*-gap) (12 lags)

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Bauer-Swanson surprise | none | 12 | 369 | −0.29 | −0.07 | 0.03 | −0.21** | −0.44* | −0.69** |
| | | | | [−0.68, 0.09] | [−0.48, 0.25] | [−0.41, 0.34] | [−0.36, −0.04] | [−0.71, −0.09] | [−1.12, −0.17] |
| | r*-gap | 12 | 369 | 0.40 | 0.54** | 0.47 | −0.44** | −0.84** | −1.15** |
| | | | | [−0.12, 0.90] | [0.08, 1.11] | [−0.01, 1.14] | [−0.64, −0.21] | [−1.29, −0.36] | [−1.96, −0.41] |
| Jarocinski-Karadi surprise | none | 12 | 369 | −0.29 | −0.07 | 0.03 | −0.21** | −0.44* | −0.68** |
| | | | | [−0.69, 0.10] | [−0.48, 0.24] | [−0.41, 0.34] | [−0.36, −0.04] | [−0.71, −0.09] | [−1.12, −0.17] |
| | r*-gap | 12 | 369 | 0.42 | 0.56** | 0.49* | −0.45** | −0.85** | −1.17** |
| | | | | [−0.07, 0.91] | [0.09, 1.13] | [0.00, 1.13] | [−0.63, −0.22] | [−1.30, −0.38] | [−1.91, −0.42] |

Bauer-Swanson: orthogonalized surprise of Bauer and Swanson (2023), announcements through 2019. Jarocinski-Karadi: Nakamura-Steinsson surprise kept when the 2-year yield and the SP 500 move in opposite directions in the window. The window impact and the bootstrap use the events with a surprise. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A7:** Baseline: 1996 to the present, all data in, no trend

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | none | 3 | 369 | −0.59** | −0.45** | −0.36** | 0.11 | 0.08 | 0.05 |
| | | | | [−0.83, −0.30] | [−0.74, −0.15] | [−0.69, −0.06] | [−0.06, 0.30] | [−0.19, 0.42] | [−0.30, 0.52] |
| | | 6 | 369 | −0.55** | −0.42** | −0.30 | 0.07 | 0.01 | −0.10 |
| | | | | [−0.79, −0.24] | [−0.70, −0.09] | [−0.63, 0.01] | [−0.11, 0.27] | [−0.28, 0.33] | [−0.47, 0.39] |
| | | 12 | 369 | −0.81** | −0.49** | −0.31 | 0.09 | 0.02 | −0.12 |
| | | | | [−1.22, −0.42] | [−0.80, −0.13] | [−0.65, 0.06] | [−0.09, 0.31] | [−0.33, 0.44] | [−0.62, 0.46] |
| | nominal | 3 | 369 | −0.02 | 0.07 | 0.05 | −0.10 | −0.27 | −0.37 |
| | | | | [−0.35, 0.33] | [−0.30, 0.42] | [−0.32, 0.37] | [−0.31, 0.08] | [−0.61, 0.08] | [−0.81, 0.09] |
| | | 6 | 369 | 0.01 | 0.17 | 0.17 | −0.14 | −0.39* | −0.60** |
| | | | | [−0.33, 0.32] | [−0.24, 0.53] | [−0.24, 0.56] | [−0.35, 0.07] | [−0.74, −0.01] | [−1.11, −0.09] |
| | | 12 | 369 | −0.05 | 0.16 | 0.22 | −0.13 | −0.43* | −0.71* |
| | | | | [−0.53, 0.37] | [−0.27, 0.54] | [−0.18, 0.62] | [−0.32, 0.08] | [−0.83, −0.02] | [−1.30, −0.07] |
| Real | none | 3 | 369 | −0.34** | −0.33** | −0.29** | −0.05 | −0.07 | −0.09 |
| | | | | [−0.56, −0.16] | [−0.57, −0.10] | [−0.56, −0.05] | [−0.19, 0.09] | [−0.31, 0.17] | [−0.41, 0.26] |
| | | 6 | 369 | −0.33** | −0.29* | −0.22 | −0.01 | −0.09 | −0.19 |
| | | | | [−0.59, −0.11] | [−0.57, −0.03] | [−0.53, 0.06] | [−0.16, 0.16] | [−0.35, 0.20] | [−0.57, 0.23] |
| | | 12 | 369 | −0.60** | −0.33** | −0.22 | −0.03 | −0.11 | −0.21 |
| | | | | [−0.94, −0.24] | [−0.63, −0.05] | [−0.51, 0.07] | [−0.18, 0.17] | [−0.36, 0.23] | [−0.58, 0.26] |
| | real | 3 | 369 | 0.12 | 0.06 | 0.00 | −0.20 | −0.31 | −0.37 |
| | | | | [−0.17, 0.35] | [−0.25, 0.33] | [−0.27, 0.31] | [−0.38, 0.00] | [−0.58, 0.04] | [−0.71, 0.09] |
| | | 6 | 369 | 0.14 | 0.19 | 0.14 | −0.20* | −0.42** | −0.60** |
| | | | | [−0.17, 0.42] | [−0.15, 0.49] | [−0.20, 0.46] | [−0.39, −0.02] | [−0.78, −0.09] | [−1.09, −0.12] |
| | | 12 | 369 | 0.09 | 0.23 | 0.21 | −0.23** | −0.48** | −0.66** |
| | | | | [−0.30, 0.46] | [−0.18, 0.60] | [−0.15, 0.62] | [−0.43, −0.06] | [−0.86, −0.12] | [−1.26, −0.10] |
| r*-gap | none | 3 | 369 | −0.17 | −0.16 | −0.14 | −0.12 | −0.15 | −0.17 |
| | | | | [−0.47, 0.11] | [−0.52, 0.15] | [−0.48, 0.14] | [−0.27, 0.07] | [−0.42, 0.22] | [−0.54, 0.34] |
| | | 6 | 369 | −0.09 | −0.06 | −0.01 | −0.16 | −0.35* | −0.53* |
| | | | | [−0.41, 0.22] | [−0.42, 0.28] | [−0.37, 0.28] | [−0.31, 0.02] | [−0.64, −0.01] | [−0.93, −0.01] |
| | | 12 | 369 | −0.29 | −0.07 | 0.03 | −0.21** | −0.44* | −0.69** |
| | | | | [−0.68, 0.10] | [−0.48, 0.25] | [−0.41, 0.34] | [−0.36, −0.04] | [−0.71, −0.09] | [−1.12, −0.17] |
| | r*-gap | 3 | 369 | 0.26 | 0.13 | 0.01 | −0.20 | −0.25 | −0.27 |
| | | | | [−0.09, 0.60] | [−0.27, 0.48] | [−0.41, 0.39] | [−0.40, 0.04] | [−0.64, 0.17] | [−0.78, 0.26] |
| | | 6 | 369 | 0.34 | 0.36 | 0.26 | −0.33** | −0.63** | −0.87** |
| | | | | [−0.01, 0.72] | [−0.09, 0.79] | [−0.18, 0.67] | [−0.53, −0.09] | [−1.04, −0.15] | [−1.49, −0.18] |
| | | 12 | 369 | 0.34 | 0.50* | 0.43 | −0.42** | −0.81** | −1.11** |
| | | | | [−0.15, 0.79] | [0.06, 1.02] | [−0.03, 1.03] | [−0.58, −0.20] | [−1.19, −0.35] | [−1.77, −0.40] |

Nominal, real and r*-gap 2-year yield; the FCI is the matching index of Note A in each case. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A8:** Robustness 1a: monthly equations start in 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 642 | 0.01 | 0.05 | 0.05 | −0.09 | −0.15 | −0.20 |
| | | | | [−0.09, 0.13] | [−0.09, 0.18] | [−0.08, 0.19] | [−0.11, 0.17] | [−0.17, 0.45] | [−0.22, 0.73] |
| | | 6 | 639 | 0.00 | 0.03 | 0.03 | −0.12 | −0.31 | −0.53 |
| | | | | [−0.13, 0.12] | [−0.13, 0.18] | [−0.11, 0.16] | [−0.20, 0.05] | [−0.49, 0.11] | [−0.77, 0.22] |
| | | 12 | 633 | −0.05 | −0.01 | 0.00 | −0.22** | −0.50** | −0.79** |
| | | | | [−0.21, 0.09] | [−0.16, 0.13] | [−0.16, 0.13] | [−0.32, −0.09] | [−0.70, −0.17] | [−1.11, −0.26] |
| | r*-gap | 3 | 642 | 0.06 | 0.00 | −0.04 | −0.08 | −0.16 | −0.27 |
| | | | | [−0.14, 0.16] | [−0.28, 0.13] | [−0.31, 0.10] | [−0.12, 0.16] | [−0.26, 0.38] | [−0.52, 0.53] |
| | | 6 | 639 | 0.09 | 0.05 | 0.05 | −0.14 | −0.35 | −0.61 |
| | | | | [−0.10, 0.24] | [−0.24, 0.23] | [−0.26, 0.22] | [−0.23, 0.05] | [−0.54, 0.12] | [−0.90, 0.15] |
| | | 12 | 633 | 0.18 | 0.14 | 0.13 | −0.27** | −0.63** | −0.98** |
| | | | | [−0.08, 0.38] | [−0.15, 0.35] | [−0.17, 0.35] | [−0.38, −0.10] | [−0.83, −0.24] | [−1.32, −0.30] |

r*-gap only. Pre-1990 months use monthly series in place of the daily data (see lib/model.py); shocks are the 252 announcements of 1996 on. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A9:** Robustness 1b: 2020 dropped from all equations, start 1996

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 359 | 0.13 | 0.22 | 0.26 | −0.18* | −0.28 | −0.35 |
| | | | | [−0.06, 0.29] | [−0.11, 0.49] | [−0.13, 0.58] | [−0.33, 0.00] | [−0.55, 0.03] | [−0.69, 0.06] |
| | | 6 | 359 | 0.04 | 0.05 | 0.07 | −0.15 | −0.30 | −0.44 |
| | | | | [−0.14, 0.20] | [−0.24, 0.29] | [−0.27, 0.39] | [−0.29, 0.02] | [−0.56, 0.03] | [−0.81, 0.08] |
| | | 12 | 359 | −0.10 | −0.21 | −0.25 | −0.20** | −0.32* | −0.44 |
| | | | | [−0.28, 0.08] | [−0.48, 0.08] | [−0.59, 0.12] | [−0.33, −0.04] | [−0.60, −0.01] | [−0.89, 0.03] |
| | r*-gap | 3 | 359 | 0.37** | 0.44** | 0.39* | −0.22* | −0.35 | −0.47 |
| | | | | [0.16, 0.57] | [0.13, 0.78] | [0.01, 0.81] | [−0.43, −0.01] | [−0.71, 0.03] | [−0.96, 0.04] |
| | | 6 | 359 | 0.38** | 0.44* | 0.43* | −0.30** | −0.57** | −0.80* |
| | | | | [0.17, 0.56] | [0.09, 0.72] | [0.01, 0.77] | [−0.47, −0.08] | [−0.93, −0.10] | [−1.35, −0.07] |
| | | 12 | 359 | 0.41** | 0.45** | 0.40 | −0.38** | −0.69** | −0.94** |
| | | | | [0.20, 0.58] | [0.08, 0.73] | [−0.02, 0.79] | [−0.52, −0.17] | [−1.04, −0.24] | [−1.49, −0.20] |

r*-gap only. March to December 2020 dropped from the monthly and the daily equations. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A10:** Robustness 1c: 2020 dropped from all equations, start 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 632 | 0.06 | 0.11 | 0.13 | −0.11 | −0.20 | −0.27 |
| | | | | [−0.03, 0.15] | [−0.03, 0.26] | [−0.03, 0.29] | [−0.14, 0.15] | [−0.23, 0.38] | [−0.30, 0.64] |
| | | 6 | 629 | 0.11 | 0.10 | 0.05 | −0.16* | −0.39 | −0.62 |
| | | | | [−0.01, 0.18] | [−0.08, 0.24] | [−0.13, 0.20] | [−0.24, 0.00] | [−0.51, 0.02] | [−0.80, 0.08] |
| | | 12 | 623 | 0.17** | 0.15 | 0.05 | −0.28** | −0.64** | −0.97** |
| | | | | [0.05, 0.29] | [−0.02, 0.29] | [−0.13, 0.22] | [−0.37, −0.15] | [−0.82, −0.32] | [−1.31, −0.45] |
| | r*-gap | 3 | 632 | 0.12 | 0.09 | 0.06 | −0.09 | −0.25 | −0.48 |
| | | | | [−0.01, 0.20] | [−0.09, 0.21] | [−0.14, 0.21] | [−0.14, 0.16] | [−0.38, 0.26] | [−0.69, 0.28] |
| | | 6 | 629 | 0.27** | 0.15 | 0.05 | −0.20* | −0.49** | −0.80* |
| | | | | [0.11, 0.37] | [−0.09, 0.28] | [−0.17, 0.20] | [−0.29, −0.01] | [−0.63, −0.05] | [−0.98, −0.05] |
| | | 12 | 623 | 0.33** | 0.24 | 0.15 | −0.32** | −0.76** | −1.14** |
| | | | | [0.14, 0.46] | [−0.02, 0.45] | [−0.11, 0.39] | [−0.44, −0.16] | [−1.00, −0.38] | [−1.57, −0.50] |

r*-gap only. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A11:** Robustness 1d: sample ends in December 2019, start 1996

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 288 | −0.06 | −0.12 | −0.14 | −0.03 | 0.01 | 0.03 |
| | | | | [−0.29, 0.22] | [−0.49, 0.32] | [−0.57, 0.33] | [−0.13, 0.07] | [−0.16, 0.17] | [−0.18, 0.23] |
| | | 6 | 288 | −0.09 | −0.17 | −0.19 | 0.00 | 0.01 | 0.02 |
| | | | | [−0.33, 0.14] | [−0.56, 0.23] | [−0.64, 0.24] | [−0.10, 0.10] | [−0.14, 0.18] | [−0.18, 0.25] |
| | | 12 | 288 | −0.15 | −0.24 | −0.26 | 0.08 | 0.06 | 0.03 |
| | | | | [−0.38, 0.11] | [−0.58, 0.15] | [−0.68, 0.21] | [−0.05, 0.20] | [−0.12, 0.20] | [−0.19, 0.22] |
| | r*-gap | 3 | 288 | 0.24 | 0.19 | 0.11 | −0.11 | −0.10 | −0.08 |
| | | | | [−0.05, 0.54] | [−0.20, 0.59] | [−0.28, 0.53] | [−0.24, 0.03] | [−0.30, 0.10] | [−0.34, 0.15] |
| | | 6 | 288 | 0.23 | 0.10 | 0.02 | −0.12 | −0.10 | −0.10 |
| | | | | [−0.07, 0.47] | [−0.32, 0.54] | [−0.42, 0.52] | [−0.27, 0.04] | [−0.35, 0.13] | [−0.41, 0.15] |
| | | 12 | 288 | 0.30* | 0.13 | 0.00 | −0.08 | −0.14 | −0.21 |
| | | | | [0.02, 0.49] | [−0.24, 0.48] | [−0.45, 0.44] | [−0.24, 0.06] | [−0.41, 0.09] | [−0.55, 0.08] |

r*-gap only. All equations, monthly and daily. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A12:** Robustness 1e: sample ends in December 2019, start 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 561 | 0.03 | 0.05 | 0.06 | −0.04 | −0.06 | −0.07 |
| | | | | [−0.08, 0.12] | [−0.10, 0.19] | [−0.10, 0.20] | [−0.07, 0.22] | [−0.08, 0.56] | [−0.08, 0.84] |
| | | 6 | 558 | 0.11 | 0.09 | 0.04 | −0.10 | −0.26 | −0.44 |
| | | | | [−0.03, 0.18] | [−0.10, 0.20] | [−0.15, 0.15] | [−0.18, 0.07] | [−0.40, 0.19] | [−0.65, 0.33] |
| | | 12 | 552 | 0.19** | 0.18 | 0.08 | −0.21** | −0.49** | −0.82** |
| | | | | [0.04, 0.30] | [−0.03, 0.32] | [−0.15, 0.24] | [−0.32, −0.06] | [−0.71, −0.15] | [−1.18, −0.25] |
| | r*-gap | 3 | 561 | 0.04 | −0.02 | −0.04 | −0.02 | −0.13 | −0.32 |
| | | | | [−0.11, 0.12] | [−0.23, 0.10] | [−0.28, 0.09] | [−0.08, 0.19] | [−0.35, 0.32] | [−0.67, 0.35] |
| | | 6 | 558 | 0.21* | 0.04 | −0.02 | −0.12 | −0.32 | −0.56 |
| | | | | [0.01, 0.30] | [−0.22, 0.16] | [−0.27, 0.11] | [−0.22, 0.08] | [−0.51, 0.13] | [−0.86, 0.15] |
| | | 12 | 552 | 0.27** | 0.16 | 0.09 | −0.22** | −0.50** | −0.84** |
| | | | | [0.04, 0.40] | [−0.16, 0.34] | [−0.26, 0.29] | [−0.32, −0.05] | [−0.72, −0.18] | [−1.23, −0.29] |

r*-gap only. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A13:** Robustness 2a: Kiley r*, start 1996

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap (Kiley) | none | 3 | 369 | −0.33** | −0.35** | −0.31** | −0.06 | −0.04 | −0.03 |
| | | | | [−0.61, −0.12] | [−0.66, −0.09] | [−0.63, −0.05] | [−0.19, 0.11] | [−0.28, 0.24] | [−0.37, 0.36] |
| | | 6 | 369 | −0.28** | −0.28 | −0.23 | −0.07 | −0.12 | −0.19 |
| | | | | [−0.53, −0.06] | [−0.54, 0.01] | [−0.50, 0.04] | [−0.21, 0.10] | [−0.38, 0.17] | [−0.56, 0.27] |
| | | 12 | 369 | −0.53** | −0.32* | −0.23 | −0.09 | −0.10 | −0.13 |
| | | | | [−0.89, −0.19] | [−0.64, −0.03] | [−0.58, 0.06] | [−0.24, 0.07] | [−0.36, 0.23] | [−0.55, 0.36] |
| | r*-gap (Kiley) | 3 | 369 | 0.15 | −0.06 | −0.14 | −0.10 | −0.07 | −0.07 |
| | | | | [−0.16, 0.47] | [−0.36, 0.25] | [−0.38, 0.10] | [−0.31, 0.11] | [−0.44, 0.27] | [−0.51, 0.34] |
| | | 6 | 369 | 0.34* | 0.15 | 0.00 | −0.21* | −0.27 | −0.33 |
| | | | | [0.03, 0.66] | [−0.19, 0.50] | [−0.25, 0.35] | [−0.41, −0.02] | [−0.69, 0.09] | [−0.94, 0.14] |
| | | 12 | 369 | 0.39 | 0.28 | −0.01 | −0.25** | −0.18 | −0.03 |
| | | | | [0.00, 0.80] | [−0.18, 0.64] | [−0.48, 0.36] | [−0.42, −0.04] | [−0.54, 0.23] | [−0.55, 0.57] |

2-year minus expected inflation minus Kiley's r*; the FCI is Note A's index built with Kiley's r* (same adjustment in rate and FCI). Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A14:** Robustness 2b: Kiley r*, start 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap (Kiley) | none | 3 | 642 | −0.02 | 0.00 | 0.00 | −0.04 | −0.02 | 0.01* |
| | | | | [−0.12, 0.09] | [−0.11, 0.12] | [−0.10, 0.10] | [−0.06, 0.20] | [−0.01, 0.48] | [0.01, 0.74] |
| | | 6 | 639 | −0.04 | −0.01 | −0.01 | −0.07 | −0.17 | −0.28 |
| | | | | [−0.16, 0.07] | [−0.14, 0.11] | [−0.12, 0.10] | [−0.16, 0.08] | [−0.34, 0.23] | [−0.56, 0.40] |
| | | 12 | 633 | −0.11 | −0.05 | −0.03 | −0.18** | −0.35* | −0.50* |
| | | | | [−0.25, 0.04] | [−0.18, 0.08] | [−0.16, 0.09] | [−0.28, −0.03] | [−0.57, −0.05] | [−0.87, −0.01] |
| | r*-gap (Kiley) | 3 | 642 | −0.07 | −0.11 | −0.11 | −0.05 | −0.11 | −0.18 |
| | | | | [−0.25, 0.04] | [−0.31, 0.01] | [−0.28, 0.01] | [−0.10, 0.13] | [−0.24, 0.24] | [−0.40, 0.31] |
| | | 6 | 639 | −0.06 | −0.09 | −0.07 | −0.08 | −0.18 | −0.30 |
| | | | | [−0.23, 0.06] | [−0.27, 0.05] | [−0.24, 0.06] | [−0.17, 0.06] | [−0.37, 0.12] | [−0.62, 0.15] |
| | | 12 | 633 | 0.00 | −0.04 | −0.03 | −0.20** | −0.37** | −0.50** |
| | | | | [−0.20, 0.14] | [−0.24, 0.11] | [−0.21, 0.10] | [−0.29, −0.05] | [−0.55, −0.10] | [−0.81, −0.10] |

As 2a. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A15:** Robustness 2c: real 2-year and real FCI with a cubic spline trend, start 1996

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Real | none | 3 | 369 | −0.44** | −0.40** | −0.34** | −0.07 | −0.14 | −0.21 |
| | | | | [−0.74, −0.19] | [−0.74, −0.12] | [−0.71, −0.08] | [−0.21, 0.10] | [−0.41, 0.15] | [−0.58, 0.23] |
| | | 6 | 369 | −0.44** | −0.34** | −0.24 | −0.03 | −0.21 | −0.41 |
| | | | | [−0.72, −0.18] | [−0.64, −0.05] | [−0.58, 0.00] | [−0.18, 0.13] | [−0.48, 0.08] | [−0.77, 0.07] |
| | | 12 | 369 | −0.64** | −0.34* | −0.20 | 0.03 | −0.06 | −0.21 |
| | | | | [−1.04, −0.32] | [−0.68, −0.04] | [−0.52, 0.08] | [−0.11, 0.20] | [−0.33, 0.31] | [−0.61, 0.31] |
| | real | 3 | 369 | 0.00 | −0.03 | −0.05 | −0.17 | −0.29 | −0.36 |
| | | | | [−0.37, 0.30] | [−0.38, 0.22] | [−0.35, 0.20] | [−0.36, 0.01] | [−0.61, 0.01] | [−0.77, 0.03] |
| | | 6 | 369 | 0.11 | 0.20 | 0.15 | −0.20* | −0.49** | −0.68** |
| | | | | [−0.26, 0.36] | [−0.23, 0.45] | [−0.19, 0.43] | [−0.37, 0.00] | [−0.79, −0.11] | [−1.12, −0.16] |
| | | 12 | 369 | 0.19 | 0.28 | 0.17 | −0.14 | −0.25 | −0.27 |
| | | | | [−0.23, 0.54] | [−0.17, 0.63] | [−0.23, 0.53] | [−0.30, 0.06] | [−0.61, 0.12] | [−0.82, 0.27] |

No r* measure: the trend is a natural cubic spline in time (knots at the sample ends and at the NBER business-cycle peaks inside the sample), estimated in every monthly equation. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A16:** Robustness 2d: real 2-year and real FCI with a cubic spline trend, start 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Real | none | 3 | 642 | 0.01 | 0.04 | 0.04 | $-0.11^{*}$ | $-0.22^{*}$ | $-0.29^{*}$ |
| | | | | [−0.14, 0.17] | [−0.13, 0.18] | [−0.11, 0.15] | [−0.20, −0.01] | [−0.37, 0.00] | [−0.50, −0.01] |
| | | 6 | 639 | 0.01 | 0.06 | 0.06 | $-0.11^{*}$ | $-0.31^{**}$ | $-0.47^{**}$ |
| | | | | [−0.14, 0.21] | [−0.13, 0.25] | [−0.10, 0.18] | [−0.20, −0.01] | [−0.47, −0.08] | [−0.70, −0.13] |
| | | 12 | 633 | −0.10 | 0.08 | 0.11 | −0.09 | $-0.25^{**}$ | $-0.38^{**}$ |
| | | | | [−0.27, 0.06] | [−0.14, 0.27] | [−0.11, 0.24] | [−0.16, 0.01] | [−0.40, −0.05] | [−0.60, −0.07] |
| | real | 3 | 642 | 0.22 | 0.15 | 0.06 | $-0.17^{**}$ | $-0.31^{**}$ | $-0.38^{*}$ |
| | | | | [0.00, 0.41] | [−0.13, 0.37] | [−0.19, 0.30] | [−0.28, −0.04] | [−0.50, −0.06] | [−0.63, −0.06] |
| | | 6 | 639 | $0.23^{**}$ | 0.21 | 0.11 | $-0.17^{**}$ | $-0.43^{**}$ | $-0.61^{**}$ |
| | | | | [0.03, 0.42] | [−0.06, 0.40] | [−0.13, 0.32] | [−0.28, −0.04] | [−0.62, −0.13] | [−0.84, −0.18] |
| | | 12 | 633 | 0.18 | 0.18 | 0.11 | $-0.17^{**}$ | $-0.39^{**}$ | $-0.53^{**}$ |
| | | | | [−0.04, 0.38] | [−0.11, 0.44] | [−0.18, 0.34] | [−0.25, −0.04] | [−0.59, −0.14] | [−0.82, −0.14] |

As 2c; knots at the NBER peaks of 1973, 1980, 1981, 1990, 2001, 2007 and 2020. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A17:** Robustness 3a: excess bond premium as a third monthly variable, start 1996

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 369 | −0.11 | −0.14 | −0.14 | −0.14 | −0.17 | −0.19 |
| | | | | [−0.39, 0.14] | [−0.43, 0.15] | [−0.43, 0.11] | [−0.29, 0.04] | [−0.41, 0.14] | [−0.51, 0.25] |
| | | 6 | 369 | 0.02 | −0.05 | −0.09 | $-0.19^{*}$ | $-0.36^{*}$ | $-0.53^{*}$ |
| | | | | [−0.25, 0.30] | [−0.36, 0.25] | [−0.41, 0.18] | [−0.35, −0.04] | [−0.69, −0.06] | [−1.03, −0.08] |
| | | 12 | 369 | −0.08 | 0.08 | 0.16 | $-0.27^{**}$ | $-0.53^{**}$ | $-0.78^{**}$ |
| | | | | [−0.41, 0.22] | [−0.26, 0.37] | [−0.18, 0.45] | [−0.41, −0.09] | [−0.81, −0.18] | [−1.19, −0.27] |
| | r*-gap | 3 | 369 | 0.19 | 0.03 | −0.14 | −0.18 | −0.20 | −0.19 |
| | | | | [−0.16, 0.51] | [−0.41, 0.39] | [−0.54, 0.26] | [−0.38, 0.06] | [−0.51, 0.24] | [−0.59, 0.42] |
| | | 6 | 369 | 0.31 | 0.22 | 0.04 | $-0.29^{**}$ | $-0.55^{**}$ | $-0.76^{*}$ |
| | | | | [−0.02, 0.66] | [−0.23, 0.65] | [−0.39, 0.49] | [−0.47, −0.08] | [−0.93, −0.08] | [−1.30, −0.10] |
| | | 12 | 369 | 0.27 | $0.52^{*}$ | 0.42 | $-0.39^{**}$ | $-0.78^{**}$ | $-1.10^{**}$ |
| | | | | [−0.18, 0.63] | [0.02, 0.96] | [−0.07, 0.96] | [−0.55, −0.18] | [−1.15, −0.32] | [−1.75, −0.37] |

r*-gap. Monthly system u, inflation, EBP (EBP treated like u and inflation; responses shown for u and the price level). Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A18:** Robustness 3b: excess bond premium as a third monthly variable, start 1973

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| r*-gap | none | 3 | 642 | 0.00 | 0.04 | 0.04 | −0.09 | −0.15 | −0.19 |
| | | | | [−0.09, 0.12] | [−0.09, 0.17] | [−0.07, 0.16] | [−0.10, 0.16] | [−0.12, 0.46] | [−0.15, 0.71] |
| | | 6 | 639 | 0.01 | 0.03 | 0.01 | −0.12 | −0.31 | −0.51 |
| | | | | [−0.10, 0.11] | [−0.13, 0.15] | [−0.12, 0.12] | [−0.20, 0.05] | [−0.43, 0.12] | [−0.73, 0.23] |
| | | 12 | 633 | −0.03 | −0.01 | 0.01 | −0.22** | −0.50** | −0.78** |
| | | | | [−0.18, 0.08] | [−0.16, 0.09] | [−0.14, 0.13] | [−0.31, −0.09] | [−0.69, −0.16] | [−1.13, −0.23] |
| | r*-gap | 3 | 642 | 0.03 | 0.01 | −0.02 | −0.10 | −0.20 | −0.33 |
| | | | | [−0.15, 0.12] | [−0.21, 0.13] | [−0.21, 0.11] | [−0.15, 0.15] | [−0.33, 0.36] | [−0.61, 0.52] |
| | | 6 | 639 | 0.04 | 0.02 | 0.00 | −0.12 | −0.32 | −0.58 |
| | | | | [−0.14, 0.15] | [−0.20, 0.17] | [−0.20, 0.15] | [−0.22, 0.07] | [−0.53, 0.12] | [−0.96, 0.21] |
| | | 12 | 633 | 0.06 | 0.03 | 0.04 | −0.24** | −0.57** | −0.89** |
| | | | | [−0.13, 0.23] | [−0.19, 0.22] | [−0.17, 0.22] | [−0.34, −0.08] | [−0.76, −0.18] | [−1.21, −0.23] |

As 3a. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A19:** Robustness 4: the FCI replaced by its three weighted components

| | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|
| Case | FCI | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | components | 3 | 369 | −0.08 | −0.04 | −0.04 | 0.07 | 0.11 | 0.09 |
| | | | | [−0.48, 0.30] | [−0.44, 0.30] | [−0.44, 0.38] | [−0.14, 0.31] | [−0.28, 0.52] | [−0.41, 0.66] |
| | | 6 | 369 | 0.08 | 0.23 | 0.12 | −0.11 | −0.20 | −0.26 |
| | | | | [−0.31, 0.43] | [−0.26, 0.68] | [−0.34, 0.61] | [−0.33, 0.12] | [−0.61, 0.22] | [−0.85, 0.34] |
| | | 12 | 369 | 0.03 | 0.25 | 0.09 | 0.03 | 0.21 | 0.42 |
| | | | | [−0.55, 0.47] | [−0.35, 0.78] | [−0.50, 0.74] | [−0.17, 0.26] | [−0.21, 0.65] | [−0.26, 1.02] |
| Real | components | 3 | 369 | 0.09 | −0.01 | −0.06 | −0.03 | 0.00 | −0.02 |
| | | | | [−0.25, 0.36] | [−0.30, 0.28] | [−0.36, 0.29] | [−0.21, 0.17] | [−0.32, 0.30] | [−0.42, 0.34] |
| | | 6 | 369 | 0.25 | 0.21 | 0.01 | −0.11 | −0.14 | −0.14 |
| | | | | [−0.07, 0.58] | [−0.16, 0.59] | [−0.34, 0.45] | [−0.28, 0.07] | [−0.53, 0.17] | [−0.70, 0.26] |
| | | 12 | 369 | 0.17 | 0.29 | −0.03 | −0.06 | 0.14 | 0.41 |
| | | | | [−0.31, 0.56] | [−0.22, 0.69] | [−0.51, 0.50] | [−0.24, 0.15] | [−0.21, 0.52] | [−0.17, 0.86] |
| r*-gap | components | 3 | 369 | 0.05 | −0.10 | −0.16 | 0.06 | 0.23 | 0.29 |
| | | | | [−0.29, 0.36] | [−0.40, 0.19] | [−0.47, 0.14] | [−0.11, 0.27] | [−0.14, 0.57] | [−0.23, 0.77] |
| | | 6 | 369 | 0.23 | 0.12 | −0.05 | −0.13 | −0.10 | −0.07 |
| | | | | [−0.16, 0.54] | [−0.24, 0.47] | [−0.42, 0.34] | [−0.29, 0.10] | [−0.40, 0.30] | [−0.55, 0.44] |
| | | 12 | 369 | 0.20 | 0.22 | −0.09 | −0.14 | 0.06 | 0.35 |
| | | | | [−0.30, 0.60] | [−0.29, 0.70] | [−0.69, 0.43] | [−0.28, 0.04] | [−0.25, 0.40] | [−0.22, 0.95] |

The 10-year yield, equity valuation and the real dollar, each multiplied by its weight, enter the daily and monthly equations separately, in the same version as the 2-year yield; each has its own window impact. Replaces the FCI, so the same step without the FCI is the baseline row. Per +1 pp on the 2-year yield. ** and *: significant at the 5 and 10 percent levels (the 95% and 90% bootstrap intervals exclude zero); 90% intervals in brackets. Obs = monthly observations in the equations.

**Table A20:** Standard methods with a commodity price index in the first block

| | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|
| Method | Sample | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | |
| Cholesky + commodity prices, 2-year yield | 1996–2026 | 369 | −0.81* | −0.44 | −0.13 | 0.11 | 0.06 | |
| | | | [−1.24, −0.28] | [−0.83, 0.03] | [−0.45, 0.34] | [−0.12, 0.35] | [−0.36, 0.48] | [- |
| Cholesky + commodity prices, federal funds rate | 1984–2007 | 288 | −0.03 | −0.02 | −0.01 | −0.02 | −0.04 | |
| | | | [−0.07, 0.01] | [−0.06, 0.03] | [−0.04, 0.04] | [−0.09, 0.02] | [−0.13, 0.05] | [- |

Monthly data, 12 lags, constant, no trend; unemployment gap, core PCE inflation and the end-of-month policy rate (the 2-year Treasury yield, or the effective federal funds rate where stated), policy rate ordered last in the Cholesky VARs; price level = cumulated inflation/12. The log PPI for all commodities (monthly average) is added to the system and ordered before the policy rate. Unit: +1 pp on the policy rate on impact. Brackets: 90% intervals (residual bootstrap for the Cholesky VARs; exact Anderson–Rubin sets for the proxy VAR). * = excludes zero. Specification: baseline/lib/litstyle.py.

**Table A21:** Monthly proxy VAR: first-stage strength and impact on the FCI

| 2-year | FCI | $F$ (3 lags) | $F$ (6) | $F$ (12) | Proxy impact on FCI | Model impact on FCI |
|---|---|---|---|---|---|---|
| Nominal | none | 5.0 | 4.6 | 4.5 | | |
| | nominal | 2.7 | 2.1 | 1.6 | 1.73 | 1.13 |
| Real | none | 9.0 | 7.3 | 6.9 | | |
| | real | 6.0 | 4.2 | 3.0 | 1.50 | 1.03 |
| r*-gap | none | 9.1 | 8.2 | 8.0 | | |
| | r*-gap | 8.4 | 6.7 | 4.4 | 1.44 | 1.03 |

$F$: robust first-stage statistic of the policy-variable residual on the instrument (instrument months only, 1996–2026). Impact on FCI: end-of-month response of the FCI per +1 pp on the end-of-month 2-year yield, estimated from monthly residuals (proxy VAR, 12 lags) and implied by the daily model.

**Table A22:** Monthly VAR benchmark (12 lags)

| | | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-year | FCI | Identification | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | none | Proxy VAR | 12 | 369 | −0.91* | −0.61* | −0.32 | 0.03 | −0.08 | −0.26 |
| | | | | | [−1.37, −0.42] | [−1.08, −0.05] | [−0.71, 0.17] | [−1.20, 0.97] | [−1.83, 1.25] | [−2.27, 1.25] |
| | | Window impact | 12 | 369 | −0.90* | −0.63* | −0.35 | 0.13 | 0.06 | −0.09 |
| | | | | | [−1.42, −0.37] | [−1.01, −0.04] | [−0.74, 0.21] | [−0.11, 0.36] | [−0.41, 0.52] | [−0.73, 0.54] |
| | nominal | Proxy VAR | 12 | 369 | 0.72 | 0.51 | 0.32 | −0.67 | −1.02 | −1.17 |
| | | | | | unbounded | unbounded | unbounded | unbounded | unbounded | unbounded |
| | | Window impact | 12 | 369 | −0.14 | −0.09 | −0.01 | −0.13 | −0.36 | −0.57 |
| | | | | | [−0.64, 0.38] | [−0.60, 0.43] | [−0.51, 0.48] | [−0.38, 0.12] | [−0.86, 0.15] | [−1.29, 0.16] |
| Real | none | Proxy VAR | 12 | 369 | −0.72* | −0.48* | −0.22 | −0.12 | −0.29 | −0.46 |
| | | | | | [−1.29, −0.35] | [−0.99, −0.07] | [−0.62, 0.15] | [−0.97, 0.71] | [−1.50, 0.85] | [−1.85, 0.80] |
| | | Window impact | 12 | 369 | −0.65* | −0.46 | −0.23 | −0.04 | −0.16 | −0.29 |
| | | | | | [−1.12, −0.15] | [−0.94, 0.02] | [−0.66, 0.23] | [−0.30, 0.20] | [−0.66, 0.31] | [−0.97, 0.35] |
| | real | Proxy VAR | 12 | 369 | 0.72 | 0.49 | 0.30 | −0.72 | −1.05 | −1.12 |
| | | | | | [−0.21, 24.27] | [−0.33, 19.69] | [−0.25, 12.47] | [−11.42, 0.16] | [−16.08, 0.22] | [−16.57, 0.33] |
| | | Window impact | 12 | 369 | −0.02 | −0.04 | 0.00 | −0.26 | −0.49 | −0.63 |
| | | | | | [−0.61, 0.48] | [−0.54, 0.47] | [−0.49, 0.44] | [−0.57, 0.01] | [−1.00, 0.02] | [−1.30, 0.08] |
| r*-gap | none | Proxy VAR | 12 | 369 | −0.33 | −0.10 | 0.06 | −0.29 | −0.57 | −0.79 |
| | | | | | [−0.98, 0.02] | [−0.64, 0.24] | [−0.30, 0.32] | [−0.97, 0.48] | [−1.51, 0.47] | [−1.83, 0.31] |
| | | Window impact | 12 | 369 | −0.19 | −0.03 | 0.09 | −0.27* | −0.51* | −0.69* |
| | | | | | [−0.75, 0.23] | [−0.52, 0.37] | [−0.28, 0.44] | [−0.53, 0.00] | [−0.96, −0.02] | [−1.25, −0.05] |
| | r*-gap | Proxy VAR | 12 | 369 | 0.95* | 0.73 | 0.47 | −0.84* | −1.24* | −1.35* |
| | | | | | [0.01, 5.87] | [−0.04, 4.55] | [−0.05, 2.89] | [−3.17, −0.15] | [−4.39, −0.24] | [−4.51, −0.24] |
| | | Window impact | 12 | 369 | 0.33 | 0.31 | 0.25 | −0.45* | −0.77* | −0.94* |
| | | | | | [−0.25, 0.82] | [−0.25, 0.76] | [−0.23, 0.60] | [−0.71, −0.16] | [−1.26, −0.20] | [−1.54, −0.18] |

Monthly VAR in the unemployment gap, core PCE inflation, the end-of-month 2-year yield and, when included, the end-of-month FCI (same version), 1996 to September 2026, no trend, nothing dropped. Proxy VAR: Nakamura–Steinsson surprise as instrument; sets are exact 90% Anderson–Rubin sets, "unbounded" when the instrument cannot reject large effects. Window impact: impact vector taken from the daily model (the end-of-month response to a window surprise); 90% bootstrap intervals. Unit in both: +1 pp on the 2-year yield (the end-of-month response to a +1 pp window surprise is 0.92 to 1.01, so the two units are the same up to 8%). * = 90% interval excludes zero.

**Table A24:** A monthly proxy VAR when the r*-gap model is the true process

| Estimate | Data | True | Simulated median | 5th–95th percentile |
|---|---|---|---|---|
| *A. Proxy VAR with the FCI* | | | | |
| First-stage $F$ | 4.4 | | 11.1 | [2.5, 27.2] |
| Unemployment gap, 12 m | 0.95 | 0.33 | 0.35 | [−0.31, 1.26] |
| Price level, 24 m | −1.24 | −0.80 | −0.69 | [−1.91, 0.17] |
| *B. Proxy VAR without the FCI* | | | | |
| Unemployment gap, 12 m | −0.33 | 0.33 | 0.02 | [−0.70, 0.70] |
| *C. Monthly VAR with the window impact and the FCI* | | | | |
| Unemployment gap, 12 m | 0.33 | 0.34 | 0.35 | [−0.10, 0.80] |

1,000 samples of 369 months generated by the estimated r*-gap model with the FCI (12 lags), with the data's calendar and announcement days; policy reaches unemployment and prices only through the 2-year yield and the FCI. Data: the 1996 sample, 12 lags, r*-gap measures; end-of-month series in the monthly VAR. True: the model's response per +1 pp on the 2-year yield, in the window (panel C) and on the end-of-month yield (panels A and B; the window response divided by the model's end-of-month response of 0.92–1.01). Simulated: median and 5th–95th percentiles of the estimates.

# F Construction and estimation details

**Events.** I use the 252 FOMC announcements from January 1996 to July 2026 with all the window measures available in the U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). As

**Table A23:** Monthly VAR benchmark (3, 6 and 12 lags)

| | | | | | Unemployment gap (pp) | | | Price level (%) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-year | FCI | Identification | Lags | Obs | 12 m | 24 m | 36 m | 12 m | 24 m | 36 m |
| Nominal | none | Proxy VAR | 3 | 369 | −0.53 | −0.43* | −0.35* | 0.00 | −0.03 | −0.06 |
| | | | | | [−0.90, 0.06] | [−0.64, −0.10] | [−0.49, −0.13] | [−1.08, 0.77] | [−1.14, 0.76] | [−1.16, 0.72] |
| | | | 6 | 369 | −0.46 | −0.36 | −0.26 | −0.05 | −0.15 | −0.27 |
| | | | | | [−0.93, 0.36] | [−0.76, 0.34] | [−0.57, 0.27] | [−1.54, 0.85] | [−2.09, 1.02] | [−2.34, 0.98] |
| | | | 12 | 369 | −0.91* | −0.61* | −0.32 | 0.03 | −0.08 | −0.26 |
| | | | | | [−1.37, −0.42] | [−1.08, −0.05] | [−0.71, 0.17] | [−1.20, 0.97] | [−1.83, 1.25] | [−2.27, 1.25] |
| | | Window impact | 3 | 369 | −0.61* | −0.46 | −0.36 | 0.14 | 0.12 | 0.09 |
| | | | | | [−0.89, −0.13] | [−0.66, 0.05] | [−0.55, 0.09] | [−0.12, 0.29] | [−0.34, 0.34] | [−0.48, 0.38] |
| | | | 6 | 369 | −0.54* | −0.42 | −0.30 | 0.09 | 0.03 | −0.07 |
| | | | | | [−0.86, −0.16] | [−0.67, 0.05] | [−0.53, 0.11] | [−0.15, 0.27] | [−0.41, 0.31] | [−0.65, 0.31] |
| | | | 12 | 369 | −0.90* | −0.63* | −0.35 | 0.13 | 0.06 | −0.09 |
| | | | | | [−1.42, −0.37] | [−1.01, −0.04] | [−0.74, 0.21] | [−0.11, 0.36] | [−0.41, 0.52] | [−0.73, 0.54] |
| | nominal | Proxy VAR | 3 | 369 | 0.35 | 0.13 | 0.02 | −0.29 | −0.42 | −0.48 |
| | | | | | [−0.34, 173.16] | [−0.40, 100.92] | [−0.38, 62.73] | [−73.88, 0.60] | [−87.91, 0.46] | [−90.42, 0.40] |
| | | | 6 | 369 | 0.63 | 0.37 | 0.17 | −0.49 | −0.78 | −0.95 |
| | | | | | unbounded | unbounded | unbounded | unbounded | unbounded | unbounded |
| | | | 12 | 369 | 0.72 | 0.51 | 0.32 | −0.67 | −1.02 | −1.17 |
| | | | | | unbounded | unbounded | unbounded | unbounded | unbounded | unbounded |
| | | Window impact | 3 | 369 | −0.06 | 0.02 | 0.00 | −0.06 | −0.22 | −0.32 |
| | | | | | [−0.41, 0.31] | [−0.30, 0.44] | [−0.31, 0.42] | [−0.32, 0.15] | [−0.63, 0.14] | [−0.80, 0.16] |
| | | | 6 | 369 | 0.02 | 0.08 | 0.05 | −0.12 | −0.34* | −0.52* |
| | | | | | [−0.33, 0.39] | [−0.28, 0.42] | [−0.32, 0.35] | [−0.36, 0.04] | [−0.78, −0.03] | [−1.09, −0.06] |
| | | | 12 | 369 | −0.14 | −0.09 | −0.01 | −0.13 | −0.36 | −0.57 |
| | | | | | [−0.64, 0.38] | [−0.60, 0.43] | [−0.51, 0.48] | [−0.38, 0.12] | [−0.86, 0.15] | [−1.29, 0.16] |
| Real | none | Proxy VAR | 3 | 369 | −0.45* | −0.41* | −0.35* | −0.04 | −0.06 | −0.08 |
| | | | | | [−0.88, −0.13] | [−0.65, −0.22] | [−0.52, −0.22] | [−0.68, 0.62] | [−0.72, 0.62] | [−0.73, 0.59] |
| | | | 6 | 369 | −0.48* | −0.38 | −0.26 | −0.06 | −0.17 | −0.31 |
| | | | | | [−1.04, −0.05] | [−0.84, 0.03] | [−0.60, 0.06] | [−0.91, 0.76] | [−1.29, 0.89] | [−1.51, 0.82] |
| | | | 12 | 369 | −0.72* | −0.48* | −0.22 | −0.12 | −0.29 | −0.46 |
| | | | | | [−1.29, −0.35] | [−0.99, −0.07] | [−0.62, 0.15] | [−0.97, 0.71] | [−1.50, 0.85] | [−1.85, 0.80] |
| | | Window impact | 3 | 369 | −0.38* | −0.37* | −0.32 | −0.04 | −0.06 | −0.07 |
| | | | | | [−0.67, −0.05] | [−0.61, −0.01] | [−0.56, 0.01] | [−0.26, 0.14] | [−0.45, 0.23] | [−0.60, 0.29] |
| | | | 6 | 369 | −0.41 | −0.34 | −0.24 | −0.03 | −0.13 | −0.26 |
| | | | | | [−0.73, 0.01] | [−0.64, 0.14] | [−0.47, 0.19] | [−0.32, 0.17] | [−0.67, 0.19] | [−0.98, 0.17] |
| | | | 12 | 369 | −0.65* | −0.46 | −0.23 | −0.04 | −0.16 | −0.29 |
| | | | | | [−1.12, −0.15] | [−0.94, 0.02] | [−0.66, 0.23] | [−0.30, 0.20] | [−0.66, 0.31] | [−0.97, 0.35] |
| | real | Proxy VAR | 3 | 369 | 0.26 | 0.01 | −0.09 | −0.23 | −0.30 | −0.32 |
| | | | | | [−0.27, 2.00] | [−0.40, 1.23] | [−0.38, 0.77] | [−1.28, 0.48] | [−1.48, 0.40] | [−1.54, 0.37] |
| | | | 6 | 369 | 0.49 | 0.26 | 0.07 | −0.46 | −0.72 | −0.85 |
| | | | | | [−0.22, 4.80] | [−0.35, 3.58] | [−0.34, 2.24] | [−3.03, 0.44] | [−4.25, 0.40] | [−4.56, 0.31] |
| | | | 12 | 369 | 0.72 | 0.49 | 0.30 | −0.72 | −1.05 | −1.12 |
| | | | | | [−0.21, 24.27] | [−0.33, 19.69] | [−0.25, 12.47] | [−11.42, 0.16] | [−16.08, 0.22] | [−16.57, 0.33] |
| | | Window impact | 3 | 369 | 0.04 | −0.04 | −0.10 | −0.18 | −0.28 | −0.32 |
| | | | | | [−0.29, 0.38] | [−0.35, 0.36] | [−0.36, 0.28] | [−0.41, 0.04] | [−0.64, 0.10] | [−0.76, 0.16] |
| | | | 6 | 369 | 0.06 | 0.04 | −0.03 | −0.23* | −0.45* | −0.60* |
| | | | | | [−0.37, 0.43] | [−0.36, 0.46] | [−0.37, 0.37] | [−0.55, −0.02] | [−0.97, −0.07] | [−1.22, −0.07] |
| | | | 12 | 369 | −0.02 | −0.04 | 0.00 | −0.26 | −0.49 | −0.63 |
| | | | | | [−0.61, 0.48] | [−0.54, 0.47] | [−0.49, 0.44] | [−0.57, 0.01] | [−1.00, 0.02] | [−1.30, 0.08] |
| r*-gap | none | Proxy VAR | 3 | 369 | −0.27 | −0.22* | −0.17* | −0.10 | −0.13 | −0.15 |
| | | | | | [−0.80, 0.06] | [−0.51, −0.03] | [−0.34, −0.06] | [−0.71, 0.59] | [−0.75, 0.57] | [−0.77, 0.55] |
| | | | 6 | 369 | −0.21 | −0.11 | −0.03 | −0.18 | −0.38 | −0.56 |
| | | | | | [−0.86, 0.18] | [−0.61, 0.22] | [−0.36, 0.21] | [−0.88, 0.67] | [−1.28, 0.68] | [−1.49, 0.54] |
| | | | 12 | 369 | −0.33 | −0.10 | 0.06 | −0.29 | −0.57 | −0.79 |
| | | | | | [−0.98, 0.02] | [−0.64, 0.24] | [−0.30, 0.32] | [−0.97, 0.48] | [−1.51, 0.47] | [−1.83, 0.31] |
| | | Window impact | 3 | 369 | −0.16 | −0.17 | −0.14 | −0.13 | −0.15 | −0.17 |
| | | | | | [−0.56, 0.23] | [−0.52, 0.25] | [−0.46, 0.21] | [−0.37, 0.09] | [−0.58, 0.20] | [−0.71, 0.28] |
| | | | 6 | 369 | −0.06 | −0.02 | 0.03 | −0.24* | −0.45* | −0.61* |
| | | | | | [−0.62, 0.33] | [−0.48, 0.41] | [−0.34, 0.38] | [−0.56, 0.00] | [−1.05, −0.02] | [−1.28, 0.00] |
| | | | 12 | 369 | −0.19 | −0.03 | 0.09 | −0.27* | −0.51* | −0.69* |
| | | | | | [−0.75, 0.23] | [−0.52, 0.37] | [−0.28, 0.44] | [−0.53, 0.00] | [−0.96, −0.02] | [−1.25, −0.05] |
| | r*-gap | Proxy VAR | 3 | 369 | 0.33 | 0.13 | −0.01 | −0.19 | −0.24 | −0.25 |
| | | | | | [−0.15, 1.34] | [−0.26, 0.93] | [−0.30, 0.59] | [−0.89, 0.48] | [−0.95, 0.43] | [−0.98, 0.40] |
| | | | 6 | 369 | 0.67 | 0.41 | 0.16 | −0.59 | −0.87 | −1.00* |
| | | | | | [0.00, 2.51] | [−0.12, 1.84] | [−0.20, 1.11] | [−1.80, 0.12] | [−2.40, 0.02] | [−2.53, −0.09] |
| | | | 12 | 369 | 0.95* | 0.73 | 0.47 | −0.84* | −1.24* | −1.35* |
| | | | | | [0.01, 5.87] | [−0.04, 4.55] | [−0.05, 2.89] | [−3.17, −0.15] | [−4.39, −0.24] | [−4.51, −0.24] |
| | | Window impact | 3 | 369 | 0.19 | 0.06 | −0.05 | −0.19 | −0.24 | −0.26 |
| | | | | | [−0.26, 0.56] | [−0.39, 0.47] | [−0.46, 0.36] | [−0.43, 0.05] | [−0.65, 0.16] | [−0.79, 0.27] |
| | | | 6 | 369 | 0.36 | 0.22 | 0.04 | −0.41* | −0.67* | −0.81* |

in the companion note, I leave out unscheduled announcements with no change in the policy rate, and March–April 2020. The announcements of September and October 2026 are left out because too few later days are available. The DXY index used in the window covers fewer currencies than the broad dollar used in the daily and monthly index, which I treat as a measurement difference.

**From monthly to daily.** Expected inflation and $r^*$ are not available daily, and they move slowly. I take the monthly gap between $R_t^v$ in equation (8) and $y_t^2$, interpolate it linearly between the middles of adjacent months, and add it to the daily yield. The daily index is

$$F_\tau^v = w_B\, y_\tau^{10} + w_Q\, 100 \ln \mathrm{CAPE}_\tau + w_D\, 100 \ln \mathrm{DOL}_\tau + o_\tau^v, \tag{18}$$

where $\mathrm{DOL}_\tau$ is the daily nominal broad dollar. The offset $o_\tau^v$ is the monthly difference between the index in equation (4) and the average of the first three terms, interpolated in the same way. It also absorbs the difference between the real and the nominal dollar. The monthly averages of the daily series are very close to the monthly measures: the root-mean-square difference is at most 0.02 unit for the FCI and 0.04 percentage point for the two-year yield. The offsets change little from day to day, so day-to-day moves in the index come from market prices. Table A25 shows the measures used at each frequency. The daily calendar has every weekday; missing values, such as on holidays, are filled from the previous observation for up to three days. Each daily equation has 13 coefficients with the FCI; each monthly equation has $1 + 4L$ with the FCI and $1 + 3L$ without it.

**Table A25:** Measures of stocks and the dollar at each frequency

| Frequency | Stocks | Dollar |
|---|---|---|
| Window | S&P 500 | DXY index |
| Daily | CAPE | Nominal broad dollar, plus offset |
| Monthly | CAPE, or earnings-yield gap to $r^*$ | Real broad dollar |

**Tracing the effect of a surprise.** The response is computed recursively, as a deviation from a path with no surprise. Let the announcement occur on weekday $d_0$ of the announcement month, and let $h = 0, 1, \ldots, 36$ count months from then. Tildes denote responses.

1. In each month, unemployment and inflation come first: $\tilde{m}_0 = 0$, and for $h \geq 1$,

$$\tilde{m}_h = \sum_{l=1}^{L} A_l\, \tilde{m}_{h-l} + \sum_{l=1}^{L} G_l\, \tilde{x}_{h-l}^M, \tag{19}$$

   with terms dated before the announcement month equal to zero, and $\tilde{x}_h^M$ the response of the monthly average of $x$.

2. The days of the month follow. Before day $d_0$, $\tilde{x}_\tau = 0$. On day $d_0$, $\tilde{x}_\tau = b$ from equation (10). On every later day, with $\tilde{x}_{\tau-1}^{(k)}$ the average of the responses over the $k$ days through $\tau - 1$,

$$\tilde{x}_\tau = \sum_{k \in \{1,5,22,66\}} \Phi_k\, \tilde{x}_{\tau-1}^{(k)} + \Gamma_0\, \tilde{m}_{h(\tau)} + \Gamma_1\, \tilde{m}_{h(\tau)-1}, \tag{20}$$

   where $h(\tau)$ is the month of day $\tau$, counted from the announcement month.

3. The monthly average $\tilde{x}_h^M$ then enters equation (19) for later months.

Months after the announcement month have 21 weekdays. To avoid computing a path for every date, I form six groups of announcement dates by the weekday of the month, in blocks of four

days, and compute the path at each group's average day and month length. With $\omega_g$ the share of announcements in group $g$, the reported response is $\sum_g \omega_g \tilde{m}_h^{(g)}$. Because the model is linear, these responses do not depend on the constant or the trend.

**Simulation.** The process is the estimated baseline model (relative to $r^*$, with the FCI; 12 lags), on the data's calendar of weekdays from January 1996 to September 2026 (369 months) and the actual announcement days. Within each month the macro variables are set first and the days of the month follow. Daily innovations are drawn with replacement from the model's daily residuals, separately for announcement days and other days; on announcement days the policy impact $b\, s_a$ is first removed from the residual and then added back with a drawn surprise. Monthly innovations are drawn jointly from the monthly residuals. Surprises are drawn with replacement from the 252 observed surprises. The twelve months before the sample and the 70 days before it are the actual 1995 data. On each sample I estimate the proxy VAR with and without the FCI, with the instrument and instrument months as in the data, and the window-impact VAR, taking the impact on the yield and the FCI from the model.

**Bootstrap.** The draws are those of Appendix C: 500 block-bootstrap draws, from each of which I compute the response and report percentile intervals. The sets for the Phillips multiplier in Table 4 use the same draws.

# G Data sources

Table A26 lists the data. Series from FRED are identified by their mnemonics. Quarterly series are placed in the middle month of the quarter and interpolated linearly.

**Table A26:** Data sources

| Series | Frequency | Source |
|---|---|---|
| Unemployment rate | Monthly | FRED UNRATE |
| Natural rate of unemployment | Quarterly | CBO, FRED NROU |
| Core PCE price index | Monthly | FRED PCEPILFE |
| Two-year Treasury yield | Daily | FRED DGS2; before June 1976, the Gürkaynak–Sack–Wright zero-coupon yield (Gürkaynak et al., 2007) (Federal Reserve Board) |
| 10-year Treasury yield | Daily, monthly | FRED DGS10, GS10 |
| Expected inflation, 1, 2 and 10 years | Monthly | Cleveland Fed, FRED EXPINF1YR, EXPINF2YR, EXPINF10YR; before 1982, Livingston Survey (Philadelphia Fed) |
| $r^*$, Laubach–Williams | Quarterly | New York Fed, one-sided estimates (data through 2026Q2) |
| $r^*$, Kiley | Quarterly | One-sided estimates of the author |
| CAPE | Monthly | Shiller, spliced over 1990–92 to the monthly average of daily CAPE |
| Daily CAPE | Daily | S&P 500 (FRED SP500) over Shiller's 10-year earnings, constructed by the daily workflow of the repository |
| Real broad dollar | Monthly | FRED RTWEXBGS, spliced to TWEXBPA in 2006 |
| Nominal broad dollar | Daily | FRED DTWEXBGS; earlier DTWEXB, TWEXBMTH |
| Excess bond premium | Monthly | Gilchrist and Zakrajšek (2012), Federal Reserve Board |
| Federal funds rate; commodity prices | Daily; monthly | FRED DFF; FRED PPIACO (producer price index, all commodities) |
| Futures, yields, S&P 500 and DXY in the window | Event | U.S. Monetary Policy Event-Study Database, "monetary events" (Acosta et al., 2025) |
| Orthogonalized surprise | Event | Bauer and Swanson (2023a), posted data (announcements through 2019) |